
PREDICTION DYNAMICS IN DEPTH-RECURRENT LANGUAGE MODELS

Xinyue Luo Fei Yu
Ant Group
mangduo.lxy@antgroup.com

ABSTRACT

Depth-recurrent language models refine predictions through repeated latent updates. Why can intermediate answers agree with the endpoint while their scores continue to change? We derive a sharp margin characterization that decomposes the conservatism of a magnitude bound into common translation, direction relative to the winner, and the pairing of each competitor’s update with its score gap. Across Huginn-3.5B and Ouro-1.4B, accounting for update direction and competitor pairing reduces the mean earliest qualifying depth by a further 22.5–34.4% of the total depth beyond translation removal under full answer-text scoring. This retrospective comparison uses completed trajectories. Substantial contributions also occur under label scoring. For shared predictive distributions, we separate common and contrast motion orthogonally and express the common component through candidate-set mass and within-set concentration. Common and contrast energies can attenuate at different rates, allowing a growing preference-change share to coexist with shrinking absolute updates. These findings explain finite-depth answer preservation through the geometry and composition of observed score changes.

1 Introduction

Additional inference-time computation can improve language-model reasoning, with its value depending on allocation across problems (Snell et al., 2025). Latent approaches place computation before discrete answer generation, through pause positions (Goyal et al., 2024), continuous intermediate states (Hao et al., 2025), or repeated applications of shared transformer blocks (Geiping et al., 2025; Zhu et al., 2025). Depth-recurrent models make the number of updates an inference-time variable. A benchmark score summarizes the outcome at one budget; intermediate predictions reveal how that outcome develops. We study how additional latent computation reshapes answer preferences and which components of this evolution affect the final prediction.

A large score update may shift all candidates together, enlarge the winner’s lead, or favor a competitor whose gap remains too large to close. Such updates can preserve the same answer despite substantial score changes. We derive an exact decomposition of the gap between a magnitude-based margin bound and the realized endpoint margin into translation, winner-directed motion, and competitor pairing. We then establish the corresponding ordering of earliest qualifying depths. Across Huginn and Ouro, direction and pairing account for substantial conservatism remaining after translation removal under both label selection and complete answer-text scoring. The empirical contribution is the size and composition of this gap along observed trajectories. We also characterize what common score motion contains. For scores from one predictive distribution, we first separate common and contrast components orthogonally, then express the common component through candidate-set mass and within-set concentration, including their interaction. Their depth trajectories show how the share of preference change can grow while its absolute magnitude falls. Together, the two analyses connect the composition of score changes to their relation with the decision boundary.

Paired scoring and precision comparisons examine how these findings depend on the readout construction and numerical arithmetic. Coordinate-permutation references characterize update alignment and joint variation, while held-out tests assess prediction. These comparisons place the descriptive findings alongside their empirical limits.

2 Related Work

Latent computation makes the amount of inference separable from parameter count. Adaptive computation time and Universal Transformers introduce iterative computation (Graves, 2016; Dehghani et al., 2019), while looped constructions establish its expressive power (Giannou et al., 2023; Saunshi et al., 2025). Huginn and Ouro scale recurrence to language modeling (Geiping et al., 2025; Zhu et al., 2025); relaxed weight sharing (Bae et al., 2025a), token-specific recursion (Bae et al., 2025b), and variable-length training (Jeddi et al., 2026) extend its architectural flexibility. Equilibrium models (Bai et al., 2019) and Jacobian regularization (Yang et al., 2026) connect recurrence to hidden-state stability. These perspectives motivate examining what successive latent updates change in the prediction, alongside how much computation they perform.

Intermediate readouts make that evolution observable. Vocabulary-space analysis (Geva et al., 2022), tuned probes and cross-layer maps (Belrose et al., 2023; Yom Din et al., 2024) examine predictions before the final layer; layer contrasts can also guide decoding (Chuang et al., 2024). Final-answer agreement in Ouro (Zhu et al., 2025) and answer convergence in explicit reasoning traces (Liu and Wang, 2025) connect intermediate predictions to their endpoints. Early-exit methods use such signals for stopping (Xin et al., 2020; Schuster et al., 2022; Jazbec et al., 2024), while self-speculative methods verify drafts with later computation (Elhoushi et al., 2024; Liu et al., 2024). The observed trajectory also depends on how an answer is scored: continuation likelihood is subject to surface-form competition (Holtzman et al., 2021), label selection requires binding answers to symbols (Robinson et al., 2023), and option order or label preferences can change the winner (Zheng et al., 2024). Our paired-format comparison connects these scoring effects to prediction dynamics across recurrent depth.

To analyze this evolution, relative comparisons provide a bridge between readout coordinates and the selected answer. Classical oscillation seminorms identify vectors differing by a constant (Gaubert and Qu, 2015); softmax shares that invariance (Gao and Pavel, 2017), and log-ratio coordinates represent relative probability information (Egozcue et al., 2003). Margin-based robustness relates changes in these comparisons to decision boundaries (Hein and Andriushchenko, 2017; Tsuzuku et al., 2018). Using translation-invariant comparisons and margin inequalities, we derive an exact decomposition of the gap between a magnitude-based bound and the realized endpoint margin, then examine its components across depth. For shared predictive distributions, mass and concentration give the common readout component a complementary probabilistic interpretation.

3 The Geometry of Prediction Stability

Prediction stability depends on how score updates interact with the gaps between candidate answers. For a fixed endpoint, we characterize answer preservation through these relative changes and derive a hierarchy of margin criteria, from a magnitude bound to the exact competitor constraints. This hierarchy determines the ordering of earliest qualifying depths along any completed score trajectory with a fixed candidate set.

3.1 From intermediate predictions to answer comparisons

For an input x , a recurrent model produces hidden states $h_t = \Phi(h_{t-1}; x)$ and intermediate predictive distributions through its output head. Fix an endpoint depth $T \geq 1$ and a question with $K \geq 2$ candidate answers. For scores $s_t \in \mathbb{R}^K$, let $y_t = \arg \max_k s_{t,k}$ under a deterministic tie rule. Endpoint agreement compares y_t with y_T ; accuracy compares y_t with the ground-truth label y^* . Candidate identities remain fixed across depths.

For a fixed intermediate depth, write $\delta = s_T - s_t$. Write $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^K$ for the all-ones vector.

The comparison between answers k and l changes by $\delta_k - \delta_l$. Consequently, its largest absolute change is

$$B_q := \max_{k,l} |\delta_k - \delta_l| = \text{osc}(\delta) := \max_k \delta_k - \min_k \delta_k. \quad (1)$$

All comparisons are unchanged precisely on $\text{span}\{\mathbf{1}\}$. The equivalence class $[\delta] = \delta + \text{span}\{\mathbf{1}\}$ therefore determines all comparison changes. The classical oscillation seminorm (Gaubert and Qu, 2015) induces a norm on this quotient, measuring the largest such change.

Proposition 3.1 (Comparison quotient and optimal centers). *For $K \geq 2$ and $\delta \in \mathbb{R}^K$, the map $[\delta] \mapsto \text{osc}(\delta)$ is a norm on $\mathbb{R}^K / \text{span}\{\mathbf{1}\}$. Let $\text{mid}(\delta) = (\max_k \delta_k + \min_k \delta_k)/2$. For every $b \in \mathbb{R}$,*

$$2\|\delta - b\mathbf{1}\|_\infty = B_q + 2|\text{mid}(\delta) - b|, \quad B_q = 2 \min_b \|\delta - b\mathbf{1}\|_\infty. \quad (2)$$

The midrange is optimal for the maximum-coordinate norm; mean centering, $b = \bar{\delta} = K^{-1} \sum_k \delta_k$, is optimal for squared Euclidean error. Both minimizing centers are unique.

The optimized equality is the finite-dimensional quotient-norm identity of [Gaubert and Qu \(2015, Lemma 4.1\)](#). Equation 2 gives each center's exact excess over pairwise change, including the raw radius $B_{\text{raw}} = 2\|\delta\|_\infty$ at $b = 0$. For comparison with conventional centers, define

$$b_{\text{mean}} = \bar{\delta}, \quad b_{\text{LSE}} = \text{LSE}(s_T) - \text{LSE}(s_t), \quad \text{LSE}(v) = \log \sum_{k=1}^K e^{v_k}.$$

Proof of Proposition 3.1. Let $K \geq 2$, $v \in \mathbb{R}^K$, and $\alpha = \max_k v_k$, $\beta = \min_k v_k$, $q = \alpha - \beta$, $c = (\alpha + \beta)/2$. For every $b \in \mathbb{R}$,

$$\begin{aligned} \|v - b\mathbf{1}\|_\infty &= \max\{|\alpha - b|, |\beta - b|\} \\ &= \max\{|c - b + q/2|, |c - b - q/2|\} = q/2 + |c - b|. \end{aligned} \quad (3)$$

Consequently,

$$\arg \min_{b \in \mathbb{R}} \|v - b\mathbf{1}\|_\infty = \{c\}, \quad \inf_{b \in \mathbb{R}} \|v - b\mathbf{1}\|_\infty = \frac{\text{osc}(v)}{2}.$$

For $v, w \in \mathbb{R}^K$ and $\lambda, b \in \mathbb{R}$,

$$\begin{aligned} \text{osc}(v + b\mathbf{1}) &= \text{osc}(v), & \text{osc}(\lambda v) &= |\lambda| \text{osc}(v), \\ \text{osc}(v + w) &= \max_{k,l} \{(v_k - v_l) + (w_k - w_l)\} \leq \text{osc}(v) + \text{osc}(w), \\ \text{osc}(v) &= 0 \iff v \in \text{span}\{\mathbf{1}\}. \end{aligned}$$

Thus $[v] \mapsto \text{osc}(v)$ is a norm on $\mathbb{R}^K / \text{span}\{\mathbf{1}\}$, and

$$\|[v]\|_{\infty, \text{quot}} := \inf_{b \in \mathbb{R}} \|v - b\mathbf{1}\|_\infty = \frac{1}{2} \text{osc}(v).$$

This is the finite-dimensional specialization of [Gaubert and Qu \(2015, Lemma 4.1\)](#).

For $\bar{v} = K^{-1} \sum_k v_k$, orthogonality gives

$$\begin{aligned} \|v - b\mathbf{1}\|_2^2 &= \|v - \bar{v}\mathbf{1}\|_2^2 + K(b - \bar{v})^2, \\ \arg \min_{b \in \mathbb{R}} \|v - b\mathbf{1}\|_2^2 &= \{\bar{v}\}. \end{aligned}$$

For $K = 2$, $\bar{v} = c$. For each $K \geq 3$, the examples

v	$2\ v\ _\infty$	$2\ v - \bar{v}\mathbf{1}\ _\infty$
$\mathbf{1}$	2	0
$(-1, 1, \dots, 1)$	2	$4(K-1)/K > 2$

exclude either uniform ordering of the raw and mean-centered radii. Both are at least $\text{osc}(v)$ by Equation 3. □

3.2 From comparison changes to answer stability

Fix a depth with a unique winner $a = y_t$. For each competitor $b \neq a$, define its initial gap and its relative update toward the winner as

$$g_b = s_{t,a} - s_{t,b} > 0, \quad u_b = \delta_b - \delta_a, \quad m = \min_{b \neq a} g_b, \quad h = \max_{b \neq a} u_b. \quad (4)$$

The decision region for a is the intersection of halfspaces $\{\varepsilon : \varepsilon_b - \varepsilon_a < g_b \text{ for all } b \neq a\}$. Its endpoint reserve is

$$R = \min_{b \neq a} (g_b - u_b) = \min_{b \neq a} (s_{T,a} - s_{T,b}). \quad (5)$$

Thus $R > 0$ characterizes a as the strict endpoint winner, while $R = 0$ places the endpoint on a decision boundary. The signed quantity h records the largest competitor-relative update; $h < 0$ means that every gap has increased. Margin-based perturbation analysis relates these halfspaces to update bounds ([Hein and Andriushchenko, 2017](#); [Tsuzuku et al., 2018](#)). The following result connects the sharp oscillation bound to the margin attained by the actual update.

Theorem 3.2 (Sharp comparison geometry and endpoint margin). *Fix $s_t \in \mathbb{R}^K$, $K \geq 2$, with unique winner a and gaps as in Equation 4. For every $r \geq 0$,*

$$\min_{\text{osc}(\varepsilon) \leq r} \min_{b \neq a} (g_b - \varepsilon_b + \varepsilon_a) = m - r. \quad (6)$$

Thus every allowed update preserves a as a strict winner if and only if $m > r$. For the realized update $\delta = s_T - s_t$,

$$m - B_{\text{raw}} \leq m - B_q \leq m - h \leq R, \quad (7)$$

and the total slack satisfies

$$R - (m - B_{\text{raw}}) = \underbrace{|\max_k \delta_k + \min_k \delta_k|}_{A_{\text{tr}}} + \underbrace{(B_q - h)}_{A_{\text{dir}}} + \underbrace{(R - m + h)}_{A_{\text{pair}}}. \quad (8)$$

All three terms are nonnegative, and

$$A_{\text{pair}} = \min_{b \neq a} [(g_b - m) + (h - u_b)]. \quad (9)$$

Equality $A_{\text{pair}} = 0$ holds exactly when one competitor attains both $g_b = m$ and $u_b = h$.

The three slacks isolate comparison-invariant translation, the use of a symmetric radius in place of a signed winner-directed update, and extrema attained by different competitors. Their sum is the exact gap between the raw lower bound and the realized endpoint reserve. At $r = m$, the sharp bound permits a tie, explaining the strict-winner condition.

Proof of Theorem 3.2. Let $I = \{1, \dots, K\} \setminus \{a\}$ and $\mathcal{B}_r = \{\varepsilon \in \mathbb{R}^K : \text{osc}(\varepsilon) \leq r\}$. For every $\varepsilon \in \mathcal{B}_r$ and $b \in I$,

$$g_b - \varepsilon_b + \varepsilon_a \geq m - \text{osc}(\varepsilon) \geq m - r.$$

Choose $b_\star \in \arg \min_{b \in I} g_b$ and set $\varepsilon^\star = re_{b_\star}$. Since $a \neq b_\star$,

$$\text{osc}(\varepsilon^\star) = r, \quad g_{b_\star} - \varepsilon_{b_\star}^\star + \varepsilon_a^\star = m - r.$$

Hence the minimum is attained, and

$$\min_{\varepsilon \in \mathcal{B}_r} \min_{b \in I} (g_b - \varepsilon_b + \varepsilon_a) = m - r.$$

In particular,

$$[\forall \varepsilon \in \mathcal{B}_r, \forall b \in I : g_b - \varepsilon_b + \varepsilon_a > 0] \iff m > r.$$

For the realized update, Equation 3 yields

$$B_{\text{raw}} - B_q = |\max_k \delta_k + \min_k \delta_k| \geq 0.$$

Moreover,

$$h = \max_{b \in I} (\delta_b - \delta_a) \leq B_q, \quad R = \min_{b \in I} (g_b - u_b) \geq m - h.$$

These inequalities give Equation 7. Its successive differences satisfy

$$\begin{aligned} R - (m - B_{\text{raw}}) &= (B_{\text{raw}} - B_q) + (B_q - h) + (R - m + h), \\ R - m + h &= \min_{b \in I} [(g_b - m) + (h - u_b)]. \end{aligned}$$

For every $b \in I$, $g_b - m \geq 0$ and $h - u_b \geq 0$. Since I is finite and nonempty,

$$\begin{aligned} A_{\text{pair}} = 0 &\iff \exists b \in I : (g_b - m) + (h - u_b) = 0 \\ &\iff \exists b \in I : g_b = m \text{ and } u_b = h. \end{aligned}$$

□

Since $m > 0$,

$$m - h > 0 \iff m - \max\{0, h\} > 0.$$

Clipping h at zero leaves the passing test unchanged; the slack identity uses the signed h .

3.3 Relating the geometry to recurrence depth

Fix $T \in \mathbb{N}_{\geq 1}$. For a candidate depth set $\mathcal{T} \subseteq \{1, \dots, T-1\}$, let D_j be the earliest depth with a unique winner at which bound $j \in \{\text{raw}, \text{q}, \text{dir}, \text{res}\}$ in Equation 7 is positive; set $D_j = T$ if none qualifies. Write $F_j(d) = \Pr(D_j \leq d)$ for its cumulative earliest-depth curve.

Corollary 3.3 (Nested earliest-depth distributions). *On the same candidate set, the four tests in Theorem 3.2 give $D_{\text{raw}} \geq D_{\text{q}} \geq D_{\text{dir}} \geq D_{\text{res}}$ on every trajectory. Consequently $F_{\text{raw}} \leq F_{\text{q}} \leq F_{\text{dir}} \leq F_{\text{res}}$, and the normalized depth area satisfies*

$$W_j = 1 - \frac{\mathbb{E}[D_j]}{T} = \frac{1}{T} \sum_{d=1}^{T-1} F_j(d). \quad (10)$$

Expectations are over complete questions, without independence across depths. Differences in W are areas between the curves. The proof also gives the weighted-sum form for sparse candidate sets.

Proof of Corollary 3.3. Fix $T \in \mathbb{N}_{\geq 1}$ and $\mathcal{T} \subseteq \{1, \dots, T-1\}$. For each trajectory, define

$$\mathcal{U} = \{t \in \mathcal{T} : s_t \text{ has a unique maximizer}\}.$$

For $t \in \mathcal{U}$, let $L_j(t)$ denote the corresponding bound in Equation 7, and set

$$\mathcal{A}_j = \{t \in \mathcal{U} : L_j(t) > 0\}, \quad D_j = \min(\mathcal{A}_j \cup \{T\}).$$

Theorem 3.2 implies

$$\mathcal{A}_{\text{raw}} \subseteq \mathcal{A}_{\text{q}} \subseteq \mathcal{A}_{\text{dir}} \subseteq \mathcal{A}_{\text{res}} \implies D_{\text{raw}} \geq D_{\text{q}} \geq D_{\text{dir}} \geq D_{\text{res}}.$$

For every $d \in \mathbb{R}$,

$$\{D_{\text{raw}} \leq d\} \subseteq \{D_{\text{q}} \leq d\} \subseteq \{D_{\text{dir}} \leq d\} \subseteq \{D_{\text{res}} \leq d\}.$$

Taking probabilities gives the CDF ordering. Since $1 \leq D_j \leq T$,

$$T - D_j = \sum_{d=1}^{T-1} \mathbf{1}\{D_j \leq d\}. \quad (11)$$

Linearity of expectation therefore gives

$$1 - \frac{\mathbb{E}[D_j]}{T} = \frac{1}{T} \sum_{d=1}^{T-1} \Pr(D_j \leq d) = \frac{1}{T} \sum_{d=1}^{T-1} F_j(d).$$

For $\mathcal{T} = \{t_1 < \dots < t_J\} \neq \emptyset$, set $t_{J+1} = T$. Since $D_j \in \mathcal{T} \cup \{T\}$,

$$T - D_j = \sum_{k=1}^J (t_{k+1} - t_k) \mathbf{1}\{D_j \leq t_k\}.$$

Taking expectations yields

$$W_j = \frac{1}{T} \sum_{k=1}^J (t_{k+1} - t_k) F_j(t_k). \quad (12)$$

If $\mathcal{T} = \emptyset$, then $D_j = T$ and $W_j = 0$, consistent with the empty-sum convention. $\square$

The terminal value $F_j(T) = 1$ includes the fallback. The depth areas compare earliest qualifying depths on a chosen grid; the slacks in Equation 8 concern a single interval.

For $j \in \{\text{mean}, \text{LSE}\}$, define D_j on the same candidate set by

$$D_j = \min(\{t \in \mathcal{T} : s_t \text{ has a unique winner, } m_t > 2\|s_T - s_t - b_{j,t}\mathbf{1}\|_\infty\} \cup \{T\}),$$

where $m_t = \min_{k \neq y_t} (s_{t,y_t} - s_{t,k})$ and $b_{j,t}$ is the corresponding center for $s_T - s_t$. We use the same definition of W_j and the trajectorywise reference $D_{\text{env}} = \min(D_{\text{mean}}, D_{\text{LSE}})$, with $W_{\text{env}} = 1 - \mathbb{E}[D_{\text{env}}]/T$. This reference selects the earlier qualifying depth separately for each completed trajectory.

4 Finite-Depth Prediction Geometry

The criteria in Section 3 can now be evaluated on completed trajectories. We begin with complete answer-text scoring, then examine matched-label trajectories and the broader archived centering comparisons. A coordinate-permutation reference tests how the observed updates align with candidate identities, and answer-transition counts distinguish endpoint agreement from accuracy against ground truth.

Our main model comparisons use Huginn-3.5B and Ouro-1.4B on ARC-Challenge (Clark et al., 2018) and MMLU (Hendrycks et al., 2021). Each result reports its scoring condition and question population. Model configurations, data selection and resampling procedures are collected in Appendix A.

Score constructions. In *label scoring* (F1), v_k is candidate k ’s distinct label token and all scores come from one predictive distribution:

$$s_{t,k}^{\text{lab}} = \log p_t(v_k \mid x). \quad (13)$$

In *answer-text scoring*, candidate k has n_k continuation tokens $v_{k,1:n_k}$ and character length ℓ_k , including its leading space:

$$s_{t,k}^{\text{txt}} = \frac{1}{\ell_k} \sum_{j=1}^{n_k} \log p_t(v_{k,j} \mid x, v_{k,<j}). \quad (14)$$

Here x is the rendered prompt for the condition being evaluated. Label scoring displays all answer options. We evaluate answer-text scoring both without the options block (F2) and with it (F3); F3 uses four balanced arrangements and maps predictions back to candidate identity. Label/F2 comparisons therefore change the prompt and continuation execution as well as the scoring formula. Within F3, every geometric criterion is evaluated on the same execution. The geometric analysis applies to all these real-valued score vectors; the shared-distribution probability identities apply to the label scores in Equation 13.

Empirical depth-area decomposition. Translation removal admits endpoint-preserving depths whenever $B_q < m \leq B_{\text{raw}}$. For N complete trajectories, its average contribution to normalized depth area is

$$\Delta_{\text{tr}} = W_q - W_{\text{raw}} = \frac{1}{NT} \sum_{i=1}^N (D_{\text{raw},i} - D_{q,i}). \quad (15)$$

The nested tests decompose the empirical depth area:

$$W_{\text{res}} - W_{\text{raw}} = \underbrace{(W_q - W_{\text{raw}})}_{\Delta_{\text{tr}}} + \underbrace{(W_{\text{dir}} - W_q)}_{\Delta_{\text{dir}}} + \underbrace{(W_{\text{res}} - W_{\text{dir}})}_{\Delta_{\text{pair}}}. \quad (16)$$

Nesting fixes the signs; magnitudes depend on the joint distribution of margins and updates. At a unique-winner depth, the two relaxations add admissible cases precisely when

$$\begin{aligned} \text{direction: } & h < m \leq B_q, \\ \text{pairing: } & m \leq h, \quad u_b < g_b \quad \text{for all } b \neq a. \end{aligned}$$

The signed test includes favorable updates $h < 0$, even for $K = 2$; pairing retains each competitor’s individual constraint.

4.1 Geometry under full answer-text scoring

Full answer-text scoring with all options displayed gives substantial contributions from both direction and pairing across the four model/task cells: $\Delta_{\text{dir}} = 9.77\text{--}22.07$ percentage points (pp) and $\Delta_{\text{pair}} = 10.94\text{--}12.70$ pp. Their combined contribution beyond translation removal is 22.46–34.38 pp (Figure 1). Each task has 32 previously inspected questions, reused from the paired-format study described in Appendix A.1. Four balanced arrangements are averaged within each question before averaging questions. Every method uses each arrangement’s own endpoint and terminal fallback, so paired increments compare the same executions. These are complete-population geometric comparisons; correctness is not used to select trajectories.

Table 1 gives the per-cell estimates and paired intervals on the quarter-depth grid. The mean/LSE envelope chooses the earlier passing depth for each completed trajectory. Here LSE is an algebraic center of four real-valued text scores; the shared-distribution mass identity applies only to the label scores in F1.

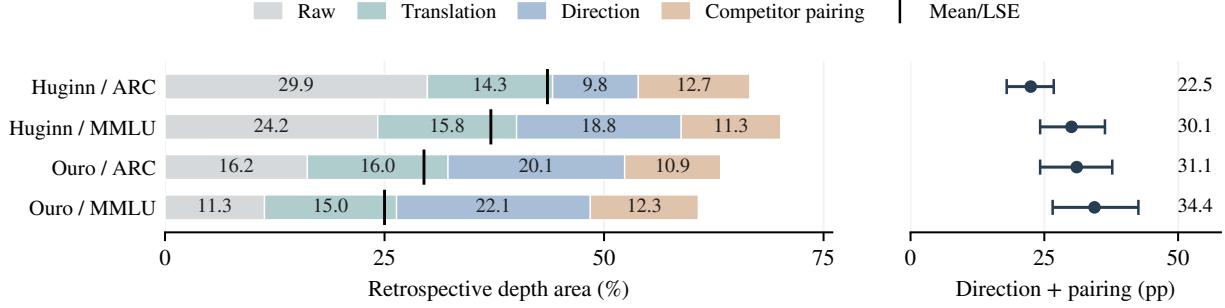


Figure 1: Direction and competitor pairing account for substantial conservatism beyond translation. Full answer-text scoring, 32 paired questions per task, four option arrangements. Left: raw depth area and the three increments in Equation 16; black marks show the mean/LSE reference. Right: the combined direction and pairing contribution, with 95% whole-question bootstrap intervals. Candidate depths are $T/4, T/2, 3T/4$ with endpoint $T = 32$ for Huginn and $T = 4$ for Ouro; $D = T$ if no earlier depth qualifies.

Table 1: Winner-aware geometry under full answer-text scoring with all options displayed. The same 32 questions per task are evaluated under four balanced arrangements in each model. Increments are retrospective depth-area percentage points on the quarter-depth grid, with marginal 95% paired question-bootstrap intervals. Native horizons are 32 for Huginn and 4 for Ouro.

Model / task	Direction	Competitor pairing	Reserve-quotient
Huginn / ARC	9.77 [7.03, 12.50]	12.70 [9.57, 16.02]	22.46 [17.97, 26.76]
Huginn / MMLU	18.75 [13.67, 24.41]	11.33 [6.84, 16.21]	30.08 [24.22, 36.33]
Ouro / ARC	20.12 [13.87, 27.15]	10.94 [6.25, 15.82]	31.05 [24.22, 37.70]
Ouro / MMLU	22.07 [15.23, 29.30]	12.30 [5.86, 19.92]	34.38 [26.56, 42.58]

Conventional centering already captures most of the translation-related improvement in these text trajectories: $W_q - W_{\text{env}} = 0.59\text{--}2.93$ pp. The quotient-over-envelope increments are 0.59 [0.00, 1.56] and 2.93 [1.37, 4.49] pp for Huginn ARC/MMLU, and 2.73 [0.98, 5.08] and 1.37 [0.20, 2.93] pp for Ouro. The larger remaining gap is resolved by the direction of relative updates and their pairing with individual gaps.

The comparison also extends across arithmetic settings and observation grids. With native output-head arithmetic, the combined direction and pairing increments are 22.46/29.69 pp for Huginn and 31.45/34.38 pp for Ouro. On Huginn’s full native candidate grid 1:31, they are 24.58/23.90 pp; Ouro’s quarter grid already includes every preterminal depth. Figure 2 shows the native-grid distributions.

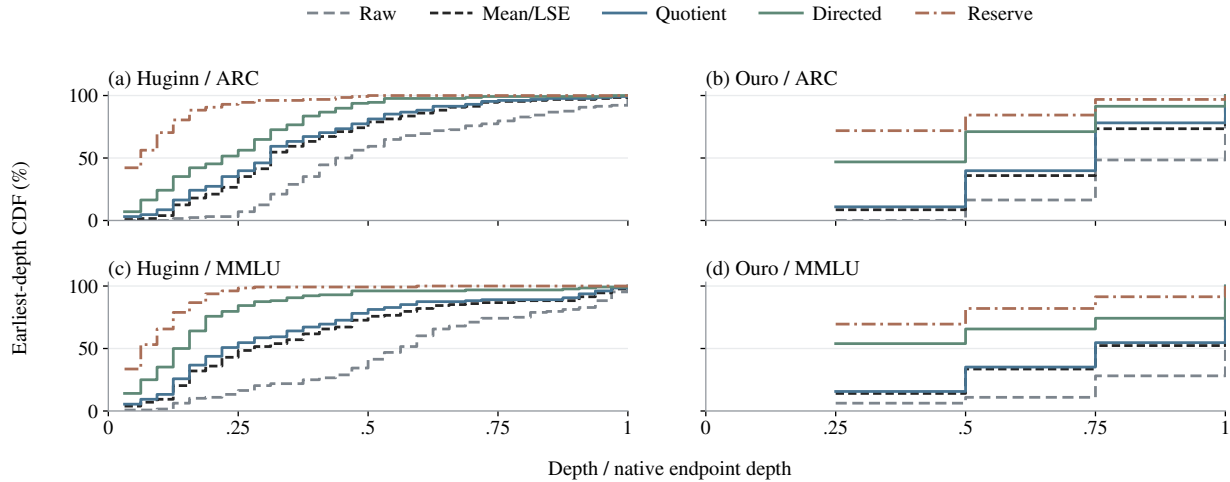


Figure 2: Earliest-depth distributions under full answer-text scoring. Curves use each model’s native candidate grid and include the shared terminal fallback. Each cell averages four arrangements within each of 32 questions. The decomposition in Figure 1 uses the quarter grid for both models.

4.2 Matched-label trajectories

The matched-label collection supplies a shared-distribution readout on a separate population of 128 questions per task, absent from the preceding question manifests. On its four model/task cells, with $\mathcal{T} = \{T/4, T/2, 3T/4\}$, $\Delta_{\text{dir}} = 8.98\text{--}22.66$ pp and $\Delta_{\text{pair}} = 4.30\text{--}10.74$ pp. Table 2 reports the absolute depth areas, and Table 3 and Figure 3 give the paired decomposition.

Table 2: Complete normalized depth areas for the four matched-label model/task conditions, in percent. Candidates are $T/4, T/2, 3T/4$; $N = 128$ per task. Passing conditions have zero endpoint disagreements.

Model	Task	T	Raw	Mean	Quotient	Directed	Reserve
Huginn-3.5B	ARC	32	22.07	29.69	32.62	43.55	54.30
	MMLU	32	32.23	37.89	42.19	51.17	61.72
Ouro-1.4B	ARC	4	31.84	35.16	37.30	51.76	56.05
	MMLU	4	21.29	23.63	26.17	48.83	55.27

Table 3: Paired increases in retrospective depth area, in percentage points, from translation, direction and competitor pairing. All four population conditions use 128 questions; intervals are paired 95% whole-question intervals.

Model	Task	$W_q - W_{\text{raw}}$	$W_{\text{dir}} - W_q$	$W_{\text{res}} - W_{\text{dir}}$
Huginn-3.5B	MMLU	9.96 [6.64, 13.48]	8.98 [6.64, 11.52]	10.55 [7.42, 13.87]
Huginn-3.5B	ARC	10.55 [7.81, 13.28]	10.94 [8.20, 13.67]	10.74 [7.42, 14.26]
Ouro-1.4B	MMLU	4.88 [3.12, 6.84]	22.66 [19.14, 26.37]	6.45 [3.91, 9.38]
Ouro-1.4B	ARC	5.47 [3.52, 7.62]	14.45 [11.72, 17.19]	4.30 [2.34, 6.64]

The quotient also improves on mean centering in every cell. The paired gains are 2.93 [1.56, 4.30] pp on Huginn ARC, 4.30 [2.34, 6.64] pp on Huginn MMLU, 2.15 [0.98, 3.52] pp on Ouro ARC, and 2.54 [1.17, 4.30] pp on Ouro MMLU.

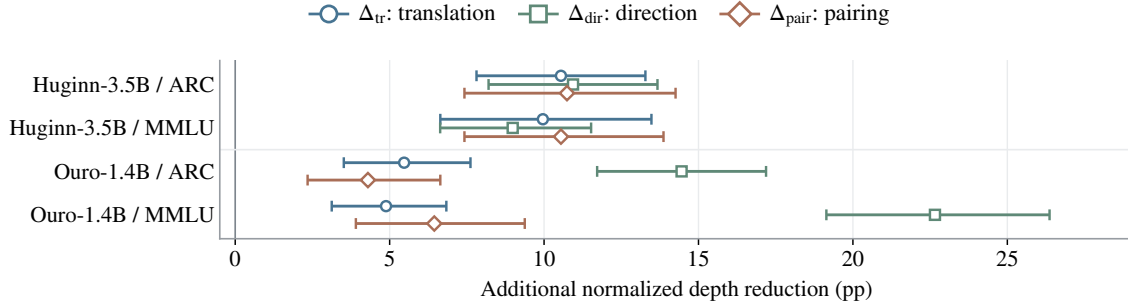


Figure 3: Direction and competitor pairing on the matched-label collections. The increments in Equation 16 on $\{T/4, T/2, 3T/4\}$, with paired 95% whole-question intervals; 128 F1 questions per model and task. Figure 4 locates the additional qualifying questions across the full native grids.

To locate the gains across recurrent depth, let $F_j(d) = N^{-1} \sum_i \mathbf{1}\{D_{j,i} \leq d\}$ on the native grid, with $N = 128$ for each label-scored cell. Nesting gives two disjoint sets of additional questions that have qualified by depth d :

$$\begin{aligned}
 G_{\text{tr}}(d) &= F_q(d) - F_{\text{raw}}(d) = \frac{1}{N} \#\{i : D_{q,i} \leq d < D_{\text{raw},i}\}, \\
 G_{\text{dp}}(d) &= F_{\text{res}}(d) - F_q(d) = \frac{1}{N} \#\{i : D_{\text{res},i} \leq d < D_{q,i}\}.
 \end{aligned} \tag{17}$$

Their sum is $F_{\text{res}}(d) - F_{\text{raw}}(d)$. On the full integer grid, $T^{-1} \sum_{d=1}^{T-1} G_{\text{tr}}(d) = W_q - W_{\text{raw}}$ and $T^{-1} \sum_{d=1}^{T-1} G_{\text{dp}}(d) = W_{\text{res}} - W_q$. Figure 4 separates these contributions at each depth. Unlike the cumulative distributions in Figure 2, these gain profiles can rise or fall as the stricter criteria catch up; both vanish at the shared terminal fallback.

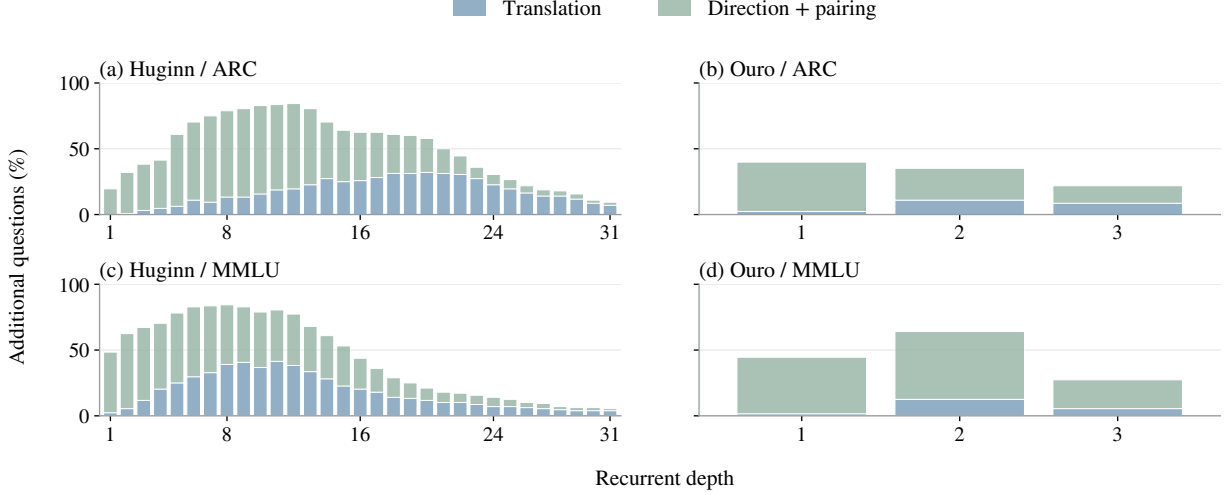


Figure 4: Where the geometric gains arise under label scoring. At each preterminal native depth d , blue gives the additional fraction of questions that have qualified after translation removal, $G_{tr}(d)$; green adds the contribution of direction and pairing, $G_{dp}(d)$. The stacked height is the reserve-over-raw gain by that depth. Each panel uses 128 F1 questions; $T = 32$ for Huginn and $T = 4$ for Ouro. Both components are zero at T , which is omitted. These profiles locate the gains, whereas Figure 2 reports complete earliest-depth distributions for the separate answer-text population.

4.3 Translation and centering on archived trajectories

The archived trajectories provide a larger population and multiple endpoints for examining translation removal and conventional centers. These previously inspected Huginn and Recurrent-Llama data retain the native evaluator’s scoring conventions: ARC uses continuation log likelihood normalized by character length, while each MMLU continuation contains one answer token. The complete population and configuration are specified in Appendix A.1. On Huginn’s $N = 2,704$ native trajectories with endpoint $T = 32$ and candidates 4:31, $W_{raw} = 42.65\%$ and $W_q = 55.28\%$, giving $\Delta_{tr} = 12.63 [12.01, 13.31]$ pp. The fraction qualifying before the endpoint changes from 88.76% to 91.46%. These statistics use endpoint displacements $s_T - s_t$. To distinguish local motion from endpoint displacement, define

$$\delta_{i,t}^{(T)} = s_{i,T} - s_{i,t}, \quad d_{i,t} = s_{i,t+1} - s_{i,t}, \quad \delta_{i,t}^{(T)} = \sum_{u=t}^{T-1} d_{i,u}.$$

Let $\mathcal{J}$ index the observed adjacent pairs in one archived collection, and set $\mathcal{I} = \{(i, t) \in \mathcal{J} : \|d_{i,t}\|_\infty > 0\}$. Assume $\mathcal{I} \neq \emptyset$. For $(i, t) \in \mathcal{I}$, define

$$r_{i,t} = 2\|d_{i,t}\|_\infty, \quad q_{i,t} = \text{osc}(d_{i,t}), \quad \rho_{i,t} = \frac{q_{i,t}}{r_{i,t}} \in [0, 1].$$

Equation 2 gives the exact identities

$$\begin{aligned} r_{i,t} - q_{i,t} &= 2|\text{mid}(d_{i,t})|, \\ \tau &:= \frac{\sum_{\mathcal{I}} (r_{i,t} - q_{i,t})}{\sum_{\mathcal{I}} r_{i,t}} = 1 - \frac{\sum_{\mathcal{I}} q_{i,t}}{\sum_{\mathcal{I}} r_{i,t}} = \sum_{\mathcal{I}} w_{i,t} (1 - \rho_{i,t}) \in [0, 1], \\ w_{i,t} &:= \frac{r_{i,t}}{\sum_{\mathcal{I}} r_{i,t}}, \quad \sum_{\mathcal{I}} w_{i,t} = 1, \\ \rho_{i,t} &= 0 \iff d_{i,t} \in \text{span}\{\mathbf{1}\} \setminus \{0\}. \end{aligned} \tag{18}$$

Here $\sum_{\mathcal{I}}$ sums over $(i, t) \in \mathcal{I}$; zero increments contribute neither radius and are omitted to avoid an undefined ratio. Thus τ measures the removable fraction of the aggregate adjacent-update radius, rather than the unweighted frequency of pure translations or a depth-area gain. In particular, $\text{osc}(\delta_{i,t}^{(T)}) \leq \sum_{u=t}^{T-1} \text{osc}(d_{i,u})$ need not be an equality, so adjacent-update ratios cannot substitute for endpoint tests. The aggregate raw-radius reductions are 44.76% in Huginn and 28.41% in Recurrent-Llama. Substantial translation-removable radius remains under FP32 arithmetic, while the exact zero-contrast increments observed in BF16 disappear (Section 6.2).

For an endpoint T and candidate set $\mathcal{T}$, let $I_t = \{i : s_{i,t} \text{ has a unique maximizer}\}$ and define

$$B_{\text{raw},i,t} = 2\|\delta_{i,t}^{(T)}\|_\infty, \quad B_{\text{q},i,t} = \text{osc}(\delta_{i,t}^{(T)}), \quad \mathcal{A}_{j,t} = \{i \in I_t : B_{j,i,t} < m_{i,t}\} \quad (j \in \{\text{raw}, \text{q}\}).$$

Since $\mathcal{A}_{\text{raw},t} \subseteq \mathcal{A}_{\text{q},t}$, the pointwise promotion set is

$$\mathcal{P}_t := \mathcal{A}_{\text{q},t} \setminus \mathcal{A}_{\text{raw},t} = \{i \in I_t : B_{\text{q},i,t} < m_{i,t} \leq B_{\text{raw},i,t}\}. \quad (19)$$

With $D_{j,i} = \min(\{t \in \mathcal{T} : i \in \mathcal{A}_{j,t}\} \cup \{T\})$,

$$D_{\text{q},i} < D_{\text{raw},i} \iff \exists t \in \mathcal{T} : i \in \mathcal{P}_t, \quad t < D_{\text{raw},i}.$$

Indeed, the forward implication takes $t = D_{\text{q},i}$; the reverse implication follows from $D_{\text{q},i} \leq t < D_{\text{raw},i}$. For Huginn ($N = 2,704$, $T = 32$), $|\mathcal{P}_{16}| = 346$. The depth-area change in Equation 15 averages the advance in earliest passing depth over complete trajectories; this pointwise count alone does not determine that area.

Mean centering gives $W_{\text{mean}} = 52.25\%$ for archived Huginn at $T = 32$. The archived trajectories provide the multi-endpoint centering comparison in Table 4. At each $T \in \{24, 28, 32\}$, all centers use the same candidates $4:T - 1$, endpoint displacement $s_T - s_t$, and terminal fallback $D = T$. For $c \in \{\text{mean}, \text{LSE}\}$, define

$$b_{\text{mean},i,t}^{(T)} = K^{-1} \mathbf{1}^\top \delta_{i,t}^{(T)}, \quad b_{\text{LSE},i,t}^{(T)} = \text{LSE}(s_{i,T}) - \text{LSE}(s_{i,t}),$$

$$B_{c,i,t} = 2\|\delta_{i,t}^{(T)} - b_{c,i,t}^{(T)} \mathbf{1}\|_\infty, \quad D_{c,i} = \min(\{t \in \mathcal{T} : i \in I_t, B_{c,i,t} < m_{i,t}\} \cup \{T\}).$$

The per-trajectory envelope and its empirical depth area are

$$D_{\text{env},i} := \min\{D_{\text{mean},i}, D_{\text{LSE},i}\}, \quad W_{\text{env}} := 1 - \frac{1}{NT} \sum_{i=1}^N D_{\text{env},i}.$$

Optimal centering implies

$$D_{\text{q},i} \leq D_{\text{env},i}, \quad W_{\text{q}} \geq W_{\text{env}} \geq \max\{W_{\text{mean}}, W_{\text{LSE}}\}.$$

The envelope therefore compares against the better center separately on each completed trajectory, which can be stronger than choosing one center for the whole population. Its selection is retrospective.

Table 4: Endpoint geometry beyond conventional centers. Paired increases in normalized depth area W (pp, 95% intervals), using all 2,704 native trajectories per checkpoint and candidates $4:T - 1$. The envelope chooses the earlier mean/LSE result separately for each trajectory.

Model	T	Quotient – mean	Quotient – LSE	Quotient – envelope
Huginn-3.5B	24	3.28 [2.89, 3.66]	3.23 [2.86, 3.62]	2.34 [2.03, 2.67]
	28	3.58 [3.18, 4.00]	3.30 [2.92, 3.70]	2.62 [2.28, 2.98]
	32	3.03 [2.67, 3.40]	2.88 [2.54, 3.24]	2.14 [1.86, 2.45]
RL-1.4B (R32)	24	0.86 [0.70, 1.04]	1.05 [0.89, 1.23]	0.62 [0.49, 0.77]
	28	0.82 [0.68, 0.97]	0.88 [0.76, 1.00]	0.54 [0.45, 0.64]
	32	0.77 [0.62, 0.93]	1.03 [0.85, 1.21]	0.60 [0.47, 0.74]

At $T = 32$, the quotient-over-envelope gains are 2.14 [1.86, 2.45] pp for Huginn and 0.60 [0.47, 0.74] pp for Recurrent-Llama.

4.4 Observed alignment and coordinate permutations

The nonnegative depth increments follow from nested criteria; they do not establish that the model’s updates align more favorably with candidate gaps than alternative assignments of the same update coordinates. To examine alignment beyond these algebraic inequalities, hold s_t fixed and enumerate all 24 coordinate permutations of each four-option δ . For $K = 4$, the exact coordinate-permutation reference is

$$p_{\text{perm}}(s_t, \delta) = \frac{1}{24} \sum_{P \in \mathfrak{S}_4} \mathbf{1}\{R_t(P\delta) > 0\}, \quad R_t(\varepsilon) = \min_{b \neq a} [g_b - (\varepsilon_b - \varepsilon_a)]. \quad (20)$$

This preserves the update’s mean, span, total energy, and contrast energy. The reference averages strict preservation over the resulting score vectors; observed preservation is then compared with it on every positive-margin question-depth pair.

Table 5 shows lower preservation in Huginn than this magnitude-matched reference and unresolved differences in Ouro. The reference acts on score coordinates; it does not impose vocabulary normalization or the model’s transition law. Independent permutations at different depths likewise supply no consistent recurrent trajectory.

Table 5: Actual preservation versus the exact coordinate-permutation reference. Values average the three quarter-grid depths within each question, then 128 questions. Differences are percentage points.

Model	Task	Actual (%)	Reference (%)	Difference [95% interval]
Huginn-3.5B	MMLU	78.12	82.34	-4.21 [-7.35, -1.12]
Huginn-3.5B	ARC	69.53	76.53	-7.00 [-10.80, -3.14]
Ouro-1.4B	MMLU	68.75	66.56	2.19 [-1.94, 6.27]
Ouro-1.4B	ARC	73.18	74.08	-0.90 [-4.29, 2.21]

4.5 Endpoint agreement and correctness

The preceding comparisons concern agreement with the fixed endpoint. Ground-truth labels distinguish the different consequences of answer changes. Let $A_t = N^{-1} \sum_i \mathbf{1}\{y_{i,t} = y_i^*\}$. For transitions $t \rightarrow T$, let $N_{W \rightarrow C}$, $N_{C \rightarrow W}$ and $N_{W \rightarrow W'}$ count corrections, degradations and switches between distinct wrong answers, respectively. Then

$$A_T - A_t = (N_{W \rightarrow C} - N_{C \rightarrow W})/N,$$

$$\frac{1}{N} \sum_i \mathbf{1}\{y_{i,t} \neq y_{i,T}\} = (N_{W \rightarrow C} + N_{C \rightarrow W} + N_{W \rightarrow W'})/N. \quad (21)$$

Endpoint disagreement therefore includes correction, degradation, and switches between wrong answers. Stability and correctness describe distinct aspects of the same trajectory.

Under full answer-text scoring, endpoint accuracy averaged over all arrangements is 49.22/43.75% for Huginn ARC/MMLU and 65.62/49.22% for Ouro. Table 6 gives the corresponding 95% question-bootstrap intervals. As in the geometric comparisons, every question and arrangement contributes to these estimates.

 Table 6: Endpoint accuracy under all-options answer-text scoring, averaged over four arrangements per question. Values are percentages with 95% whole-question intervals; $N = 32$ per task.

Model	Task	Accuracy [95% interval]
Huginn	ARC	49.22 [35.16, 62.52]
Huginn	MMLU	43.75 [28.12, 59.38]
Ouro	ARC	65.62 [50.00, 79.69]
Ouro	MMLU	49.22 [32.81, 65.62]

For the matched-label collection, label indices and choice order were matched to the original source records. Huginn’s F1 endpoint accuracy is 44/128 on MMLU and 27/128 on ARC. Table 7 separates the start-to-end transitions on all $N = 128$ questions per task. At Huginn MMLU $t = 4$, $(N_{W \rightarrow C}, N_{C \rightarrow W}, N_{W \rightarrow W'}) = (24, 7, 30)$: the net accuracy gain is 13.28 pp, while 30 answer changes leave the answer incorrect. Early MMLU contains a net correction component, while wrong-to-wrong switches remain numerous. Across the three quarter-grid depths, excess correction, degradation, and wrong-to-wrong frequencies relative to permutation are 2.04/0.37/1.80 pp for MMLU and 1.73/1.68/3.59 pp for ARC.

 Table 7: Huginn-3.5B (0125) F1 start-to-end answer transitions on 128 questions per task. $W \rightarrow W$ denotes a switch to a different wrong answer; $W=W$ denotes the same wrong answer. The five categories sum to 128 in each row. Accuracy differences are percentage points.

Task	t	$W \rightarrow C$	$C \rightarrow W$	$W \rightarrow W$	$C \rightarrow C$	$W=W$	Accuracy change [95% interval]
MMLU	1	25	10	31	19	43	11.72 [3.12, 20.31]
	4	24	7	30	20	47	13.28 [5.47, 21.88]
	8	18	12	26	26	46	4.69 [-3.91, 13.28]
	16	4	6	10	40	68	-1.56 [-6.25, 3.12]
	24	1	2	5	43	77	-0.78 [-3.12, 1.56]
	31	0	1	3	44	80	-0.78 [-2.34, 0.00]
ARC	1	22	23	58	5	20	-0.78 [-11.72, 9.38]
	4	23	27	51	4	23	-3.12 [-14.06, 7.81]
	8	21	12	44	6	45	7.03 [-1.56, 15.62]
	16	9	10	10	18	81	-0.78 [-7.03, 6.25]
	24	3	3	5	24	93	0.00 [-3.91, 3.91]
	31	2	2	1	25	98	0.00 [-3.12, 3.12]

5 Composition of Score Updates

The preceding margin analysis relates score changes to the decision boundary. Here we hold the scoring construction fixed and examine the composition of those changes. An orthogonal decomposition separates common and contrast energy; for scores from a shared predictive distribution, mass and concentration further resolve the common component. We then use this decomposition to distinguish changes in absolute energy, relative allocation, and covariance across questions.

5.1 Mass and concentration in a shared distribution

Fix a set $\mathcal{C}$ of K distinct candidate tokens with positive probabilities. For an interval (r, t) , write $\Delta f = f_t - f_r$ and $\delta = \Delta s$. Let

$$M_t = \sum_{k \in \mathcal{C}} p_{t,k}, \quad \pi_{t,k} = p_{t,k}/M_t, \quad \kappa_t = D_{\text{KL}}(U_K \| \pi_t), \quad P_{\perp} = I - \mathbf{1}\mathbf{1}^{\top}/K.$$

Here M_t is the total probability mass assigned to the candidate set, π_t is the conditional distribution within that set, and U_K is the uniform distribution on its K candidates. The reverse KL divergence κ_t measures within-set concentration relative to uniformity; $P_{\perp}$ projects onto the contrast subspace $\mathbf{1}^{\perp}$. The common Euclidean projection uses $\bar{\delta}$; the optimal L_{∞} center in Equation 2 uses the midrange.

Proposition 5.1 (Mass and concentration in a shared readout). *For selected log probabilities as in Equation 13,*

$$s_t = \log M_t \mathbf{1} + \log \pi_t. \quad (22)$$

Across any two depths, the common and contrast components satisfy

$$c := \bar{\delta} = \Delta \log M - \Delta \kappa, \quad P_{\perp} \delta = P_{\perp} \Delta \log \pi. \quad (23)$$

The common and contrast components are orthogonal, with energies

$$\begin{aligned} Q &= \|P_{\perp} \delta\|_2^2, & C &= K c^2, & E &= \|\delta\|_2^2 = Q + C, \\ C &= K(\Delta \log M)^2 + K(\Delta \kappa)^2 - 2K \Delta \log M \Delta \kappa. \end{aligned} \quad (24)$$

Thus $\delta = c\mathbf{1} \iff \Delta \pi = 0$, with $c = \Delta \log M$ in this case: a pure common update changes candidate-set mass while preserving the conditional answer distribution.

Proof of Proposition 5.1. At each of the two depths r , let $M_r = \sum_{k \in \mathcal{C}} p_{r,k}$ and $\pi_{r,k} = p_{r,k}/M_r$. Since $p_{r,k} > 0$ and $|\mathcal{C}| = K$ is fixed,

$$\begin{aligned} s_{r,k} &= \log M_r + \log \pi_{r,k}, \\ \kappa_r &= \frac{1}{K} \sum_{k \in \mathcal{C}} \log \frac{1/K}{\pi_{r,k}} = -\log K - \frac{1}{K} \sum_{k \in \mathcal{C}} \log \pi_{r,k}. \end{aligned} \quad (25)$$

Consequently,

$$\bar{s}_r = \log M_r - \log K - \kappa_r, \quad c = \bar{\delta} = \Delta \log M - \Delta \kappa.$$

Moreover, $P_{\perp} \mathbf{1} = 0$ and $P_{\perp} = P_{\perp}^{\top} = P_{\perp}^2$ imply

$$\begin{aligned} P_{\perp} \delta &= P_{\perp} \Delta \log \pi, \\ \delta &= c\mathbf{1} + P_{\perp} \delta, \quad \langle c\mathbf{1}, P_{\perp} \delta \rangle = 0. \end{aligned}$$

Hence

$$E = \|\delta\|_2^2 = K c^2 + \|P_{\perp} \delta\|_2^2 = C + Q,$$

and

$$C = K(\Delta \log M)^2 + K(\Delta \kappa)^2 - 2K \Delta \log M \Delta \kappa. \quad (26)$$

□

For vocabulary logits z , $s_{\mathcal{C}} = z_{\mathcal{C}} - \text{LSE}(z)\mathbf{1}$. Since $P_{\perp} \mathbf{1} = 0$ and oscillation is translation-invariant,

$$P_{\perp} \Delta s_{\mathcal{C}} = P_{\perp} \Delta z_{\mathcal{C}}, \quad \text{osc}(\Delta s_{\mathcal{C}}) = \text{osc}(\Delta z_{\mathcal{C}}). \quad (27)$$

Vocabulary normalization therefore leaves both contrast motion and comparison changes invariant (Gao and Pavel, 2017).

Scope of the probability interpretation. For an arbitrary score vector, $a_t = \text{LSE}(s_t)$ and $\pi_t = \text{softmax}(s_t)$ give the algebraic factorization $s_t = a_t \mathbf{1} + \log \pi_t$. It implies the same relation $\bar{\delta} = \Delta a - \Delta \kappa$. The identity becomes a statement about candidate-set probability mass when the selected entries are log probabilities from one predictive distribution. In the shared-prefix label collection this condition is built into the forward readout. For the archived MMLU precision panel it was examined through paired full-vocabulary and prefix-state comparisons, which retain finite-precision discrepancies detailed in Section 6.2. In a candidate-conditioned text score, the same algebraic a_t does not represent the mass of a single vocabulary event.

Mass and concentration both act along the common direction, so their energy contributions include a cross term. Write $E_M = K(\Delta \log M)^2$, $E_\kappa = K(\Delta \kappa)^2$ and $E_\times = -2K\Delta \log M \Delta \kappa$. Then

$$E = Q + E_M + E_\kappa + E_\times, \quad \text{sgn}(E_\times) = -\text{sgn}(\Delta \log M \Delta \kappa). \quad (28)$$

The interaction distinguishes cancellation from reinforcement within each update. Figure 5 reports mean component energies and their signed shares $\sum_i X_{i,t} / \sum_i E_{i,t}$, $X \in \{Q, E_M, E_\kappa, E_\times\}$, at every adjacent Huginn depth. The four shares sum to one. Figure 6 reports the same decomposition over Ouro’s complete native transitions, including the signed mass–concentration interaction.

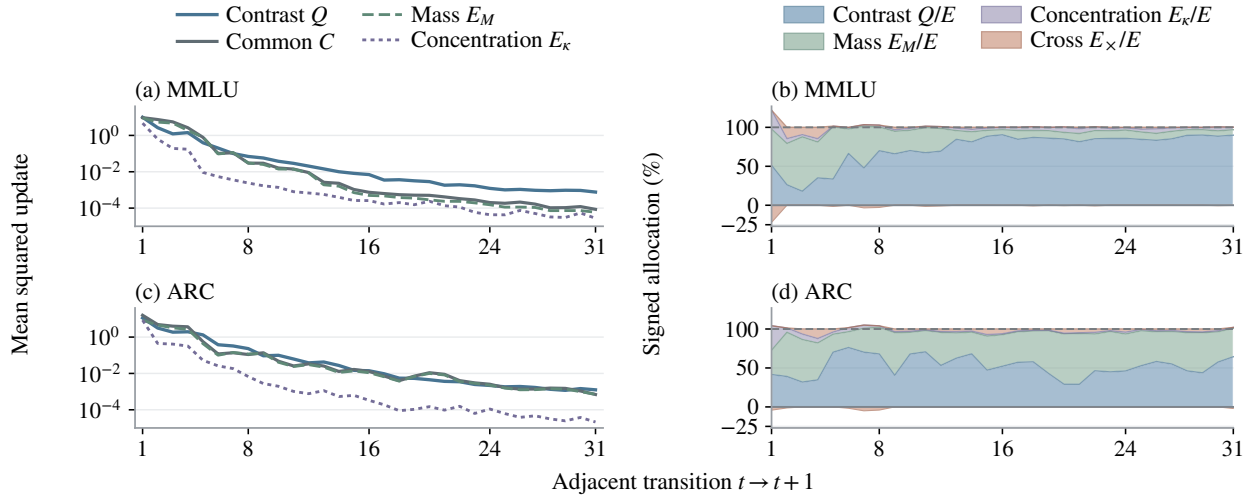


Figure 5: Changing energy allocation within contracting updates. All 31 adjacent transitions in Huginn-3.5B (0125), with 128 questions per task under label scoring. Left: mean energies on a logarithmic scale; the signed cross term appears in the right-hand panels. Right: each component’s share of total energy, with positive and negative contributions displayed separately. Late MMLU updates allocate a larger share of their energy to contrast even as contrast energy falls; ARC has a different allocation profile. These adjacent increments complement the four-step interval summaries in Table 8.

For the late Ouro MMLU interval $3 \rightarrow 4$, $(C, E_M, E_\kappa, E_\times) \approx (0.512, 0.002, 0.514, -0.004)$. Concentration dominates set-mass energy in this cell. The same common projection therefore admits different probability compositions across models and depths.

5.2 Energy attenuation and weighting

An increasing contrast share need not indicate growing contrast energy: the common component may contract faster. To compare these rates, let $Q_{i,r}, C_{i,r}$ be question i ’s energies averaged over the prescribed interval starts in window $r \in \{e, \ell\}$, and set $E_{i,r} = Q_{i,r} + C_{i,r}$. On the same N questions, write $\bar{X}_r = N^{-1} \sum_i X_{i,r}$ for $X \in \{Q, C, E\}$. We summarize contrast allocation in two ways:

$$D_{\text{pool},r} = \frac{\bar{Q}_r}{\bar{E}_r}, \quad q_{i,r} = \frac{Q_{i,r}}{E_{i,r}}, \quad D_{\text{eq},r} = \frac{1}{N} \sum_i q_{i,r}.$$

The pooled share weights questions by their total energy; the equal-question share averages their individual contrast shares. The pooled quantity requires $\bar{E}_r > 0$, and the equal-question quantity requires $E_{i,r} > 0$ for every question. If $\bar{E}_e, \bar{E}_\ell > 0$, then

$$D_{\text{pool},\ell} - D_{\text{pool},e} = \frac{\bar{Q}_\ell \bar{C}_e - \bar{Q}_e \bar{C}_\ell}{\bar{E}_\ell \bar{E}_e}. \quad (29)$$

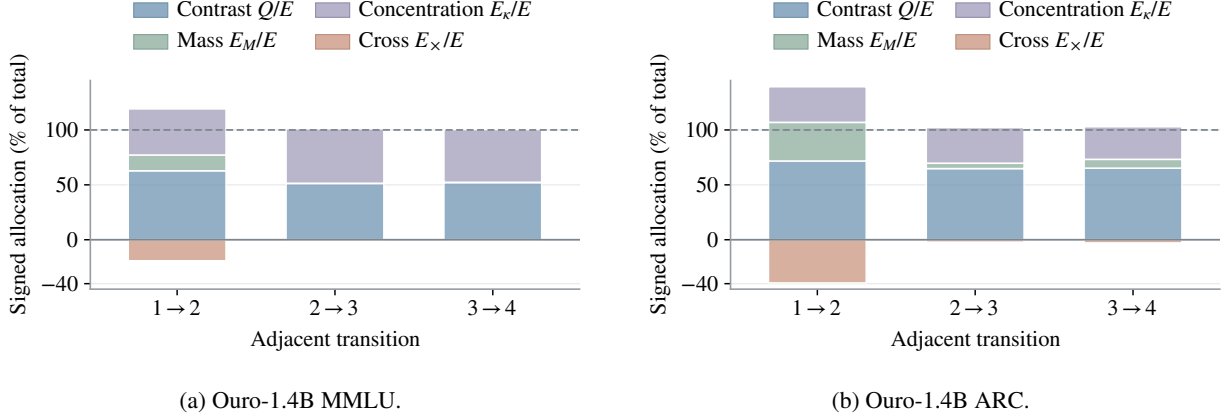


Figure 6: Signed energy allocation over Ouro-1.4B’s three native transitions, using 128 F1 questions per task. The components and denominator match Figure 5; each bar retains the signed interaction. The different native horizons describe within-model evolution and do not isolate architecture as a causal factor.

For $\bar{Q}_e, \bar{C}_e > 0$, define $\rho_Q = \bar{Q}_\ell / \bar{Q}_e$ and $\rho_C = \bar{C}_\ell / \bar{C}_e$. Consequently,

$$D_{\text{pool},\ell} > D_{\text{pool},e} \iff \rho_Q > \rho_C. \quad (30)$$

In particular, $0 \leq \rho_C < \rho_Q < 1$ gives increasing contrast share while both absolute energies decrease.

For $E_{i,r} > 0$ on every question, the difference between the two summaries is determined by their covariance under the equal-question empirical measure:

$$D_{\text{pool},r} - D_{\text{eq},r} = \frac{\text{Cov}_i(E_{i,r}, q_{i,r})}{\bar{E}_r}. \quad (31)$$

Questions with larger total energy pull the pooled share toward their own contrast allocation. For the proof, fix a window r and write $E_i = E_{i,r}$, $Q_i = Q_{i,r}$ and $d_i = q_{i,r}$. Expectations use the same equal-question empirical measure.

Proof. Under the same probability measure on questions, assume $E_i > 0$ almost surely and $0 < \mathbb{E}E_i < \infty$. Since $0 \leq d_i \leq 1$, both d_i and $E_i d_i = Q_i$ are integrable. By the definition of covariance,

$$\begin{aligned} \frac{\mathbb{E}Q_i}{\mathbb{E}E_i} &= \frac{\mathbb{E}[E_i d_i]}{\mathbb{E}E_i} \\ &= \frac{\mathbb{E}E_i \mathbb{E}d_i + \text{Cov}(E_i, d_i)}{\mathbb{E}E_i} = \mathbb{E}d_i + \frac{\text{Cov}(E_i, d_i)}{\mathbb{E}E_i}. \end{aligned}$$

□

Early and late interval comparisons. In Huginn MMLU, contrast takes a larger share in the late window even as both contrast and common energies fall. For four-step increments with starts 4:7 versus 24:27, the 128-question estimates are $D_{\text{pool},e} = 0.2170$ and $D_{\text{pool},\ell} = 0.7377$, with $\Delta D_{\text{pool}} = 0.5207$ [0.4489, 0.6010]. Table 8 gives the corresponding absolute energies and all four model/task cells. ARC’s interval includes zero.

Table 8: Absolute energies and relative allocation on 128 questions per cell under label scoring. Mean Q, C have squared-log-probability units. Huginn uses four-step increments with starts 4:7 versus 24:27; Ouro uses $1 \rightarrow 2$ versus $3 \rightarrow 4$. Both ΔD columns give late-minus-early changes. Mint/bold marks positive ΔD with lower 95% limits above zero; complete marginal intervals appear in Table 10.

Model	Task	Contrast Q		Common C		ΔD (pp)	
		Early	Late	Early	Late	Pooled	Equal
Huginn-3.5B	ARC	1.36545	.00717	1.68553	.00960	−1.98	−3.72
	MMLU	.61025	.00331	2.20178	.00118	52.07	55.54
Ouro-1.4B	ARC	4.43761	.35879	1.75132	.19187	−6.55	3.62
	MMLU	4.05283	.55714	2.40321	.51238	−10.68	5.41

The fixed depth prediction concerns both the pooled contrast share $D = D_{\text{pool}} = \mathbb{E}Q / \mathbb{E}(Q + C)$ and the mass-only approximation error $\mathcal{E}_M = \mathbb{E}[K(\bar{\delta} - \Delta \log M)^2] / \mathbb{E}C$, with positive denominators in both ratios. The latter measures

the squared error from approximating common motion by $\Delta \log M$, relative to common energy; its numerator retains the concentration contribution. Huginn compares four-step updates starting at depths 4:7 and 24:27; Ouro compares $1 \rightarrow 2$ and $3 \rightarrow 4$. The two positive MMLU changes occur on the additional Huginn questions but do not extend to Ouro under its native horizon (Table 9). Absolute Q and C decrease in all four model/task conditions. For example, Huginn/MMLU changes from $(Q, C) = (0.61025, 2.20178)$ to $(0.003312, 0.001178)$; Ouro/MMLU changes from $(4.05283, 2.40321)$ to $(0.557137, 0.512381)$.

Table 9: Original fixed early-late comparisons on 128 questions per task and model. Entries are late-minus-early differences with their original 95% intervals. D is an energy share; $\mathcal{E}_M$ is a relative squared error.

Model	Task	ΔD	$\Delta \mathcal{E}_M$
Huginn-3.5B	MMLU	0.5207 [0.4489, 0.6010]	0.2245 [0.1094, 0.3401]
Huginn-3.5B	ARC	-0.0198 [-0.1315, 0.0865]	-0.0645 [-0.1015, -0.0341]
Ouro-1.4B	MMLU	-0.1068 [-0.2104, 0.0310]	-0.1264 [-0.2752, 0.0150]
Ouro-1.4B	ARC	-0.0655 [-0.1449, 0.0186]	-0.2945 [-0.6635, 0.0696]

Question weighting. In Ouro MMLU, the pooled change is $-0.1068 [-0.2104, 0.0310]$. The equal-question change is $0.0541 [0.0054, 0.1061]$. The signs of these changes differ because the summaries assign different weights to the same questions.

The reported cells have positive aggregated energy for every question. If this condition fails, the equal-question ratio is left undefined rather than assigning a value to a zero-energy question. In Ouro/MMLU the covariance correction changes from -0.07498 to -0.23592 , reversing the sign relative to equal-question averaging. Table 10 applies both summaries to all four conditions with a common resampling implementation. Changing the observation interval also changes effect size: Huginn/MMLU pooled differences are 0.5207 for the four-step intervals, 0.4784 for matched-start one-step intervals, and 0.1866 for $8 \rightarrow 16$ versus $24 \rightarrow 32$.

Table 10: Pooled and equal-question changes in contrast share, with the same 5,000 whole-question resamples for all four conditions.

Model	Task	Pooled ΔD	Equal-question ΔD
Huginn-3.5B	MMLU	0.5207 [0.4488, 0.6014]	0.5554 [0.5137, 0.5967]
Huginn-3.5B	ARC	-0.0198 [-0.1247, 0.0837]	-0.0372 [-0.0977, 0.0219]
Ouro-1.4B	MMLU	-0.1068 [-0.2073, 0.0315]	0.0541 [0.0054, 0.1061]
Ouro-1.4B	ARC	-0.0655 [-0.1470, 0.0147]	0.0362 [-0.0365, 0.1057]

5.3 Marginal means and covariance

The common-energy share also reflects how candidate coordinates vary together across questions. For a fixed readout, let $d_{i,t} = s_{i,t+1} - s_{i,t}$ be question i 's adjacent score increment. Group these vectors by task, source subset, choice count K_g , and depth; write d_{gi} for the increment of question i in group g , which contains $n_g \geq 1$ questions. Let $P_g = \mathbf{1}\mathbf{1}^\top / K_g$ be the common projector within that group. With common energy $C_{gi} = (\mathbf{1}^\top d_{gi})^2 / K_g = \|P_g d_{gi}\|_2^2$ and total energy $E_{gi} = \|d_{gi}\|_2^2$, the observed common-energy fraction is

$$\Xi_{\text{obs}} = \frac{\sum_{g,i} C_{gi}}{\sum_{g,i} E_{gi}} = \frac{\sum_{g,i} \|P_g d_{gi}\|_2^2}{E_{\text{tot}}}, \quad E_{\text{tot}} = \sum_{g,i} E_{gi} > 0. \quad (32)$$

Write $\mu_g = \mathbb{E}_g[d]$ and $\Sigma_g = \text{Cov}_g(d)$, using equal question weights and empirical divisor n_g :

$$\mu_g = \frac{1}{n_g} \sum_i d_{gi}, \quad \Sigma_g = \frac{1}{n_g} \sum_i (d_{gi} - \mu_g)(d_{gi} - \mu_g)^\top.$$

All covariances are across questions within g .

Condition on all increment arrays. For each g, k , let σ_{gk} be independent and uniform on the permutations of $\{1, \dots, n_g\}$, and define $d_{gi,k}^\sigma = d_{g,\sigma_{gk}(i),k}$. These permutations preserve every coordinate's empirical distribution and E_{tot} exactly. Let Ξ_σ be the common-energy fraction after permutation and $\Xi_{\text{marg}} = \mathbb{E}_\sigma[\Xi_\sigma \mid \{d_{gi}\}]$ its conditional expectation. All

expectations over permutations below are conditional on these fixed arrays. Then

$$\begin{aligned}\Xi_{\text{obs}} &= \frac{\sum_g (n_g/K_g) [\mathbf{1}^\top \Sigma_g \mathbf{1} + (\mathbf{1}^\top \mu_g)^2]}{E_{\text{tot}}}, \\ \Xi_{\text{obs}} - \Xi_{\text{marg}} &= \frac{1}{E_{\text{tot}}} \sum_g \frac{n_g}{K_g} \sum_{k \neq l} \text{Cov}_g(d_k, d_l).\end{aligned}\tag{33}$$

The reference replaces $\mathbf{1}^\top \Sigma_g \mathbf{1}$ by $\text{tr}(\Sigma_g)$; the sum over $k \neq l$ uses ordered pairs. If $K_g = K$, $\mu_g = 0$, and Σ_g is diagonal for every g , then $\Xi_{\text{obs}} = \Xi_{\text{marg}} = 1/K$. Aligned nonzero means can raise Ξ_{marg} above this value.

Proof. For every g, k, l ,

$$\sum_{i=1}^{n_g} d_{gi,k} d_{gi,l} = n_g ((\Sigma_g)_{kl} + \mu_{gk} \mu_{gl}).$$

Independence and uniformity of σ_{gk} and σ_{gl} for $k \neq l$ give

$$\mathbb{E}_\sigma \left[\sum_{i=1}^{n_g} d_{gi,k}^\sigma d_{gi,l}^\sigma \right] = \begin{cases} n_g ((\Sigma_g)_{kk} + \mu_{gk}^2), & k = l, \\ n_g \mu_{gk} \mu_{gl}, & k \neq l. \end{cases}$$

Since $\|P_g d_{gi}\|_2^2 = K_g^{-1} \sum_{k,l} d_{gi,k} d_{gi,l}$ and the permutations preserve E_{tot} ,

$$\begin{aligned}\Xi_{\text{obs}} &= \frac{1}{E_{\text{tot}}} \sum_g \frac{n_g}{K_g} \{ \mathbf{1}^\top \Sigma_g \mathbf{1} + (\mathbf{1}^\top \mu_g)^2 \}, \\ \mathbb{E}_\sigma \Xi_\sigma &= \frac{1}{E_{\text{tot}}} \sum_g \frac{n_g}{K_g} \{ \text{tr}(\Sigma_g) + (\mathbf{1}^\top \mu_g)^2 \}.\end{aligned}$$

Subtracting and using $\mathbf{1}^\top \Sigma_g \mathbf{1} - \text{tr}(\Sigma_g) = \sum_{k \neq l} (\Sigma_g)_{kl}$ proves Equation 33. $\square$

The identity permits $n_g = 1$ and different K_g across groups; independence is imposed on the permutations, not on the original increments.

Across all native adjacent ARC increments, Huginn and Recurrent-Llama have common-energy fractions 77.06% and 67.51%, compared with exact marginal expectations 45.37% and 39.24%. Their covariance contributions are 31.70 and 28.27 percentage points. The complete comparisons below also report a finite-permutation estimate of the reference.

Joint variation across tasks and depths. The across-question reference preserves each coordinate’s empirical distribution within task, source subset, choice count, and depth. It answers a different question from the within-update coordinate permutations in Section 4.4.

Table 11 reports the taskwise native-precision values from 1,000 such permutations. Large common energy and positive covariance excess are different properties: Huginn MMLU has both over all increments, whereas Recurrent-Llama MMLU has a positive common fraction but a negative excess. Late MMLU, using adjacent destination depths 16:32, has observed versus reference fractions 14.38/24.83% in Huginn and 16.05/24.98% in Recurrent-Llama. At higher precision, the latter late contrast changes from $-9.56 [-12.95, -4.79]$ pp in BF16 to $-1.23 [-12.73, 9.28]$ pp in full FP32 on the 32-question panel.

Table 11: Native adjacent-increment common energy and its marginal reference on 1,352 questions per task. Values are percentages. Observed intervals are 95% whole-question intervals; reference entries are conditional permutation means. This across-question reference differs from the within-question displacement-coordinate comparison in Table 5.

Model	Task	Observed Ξ [95% interval]	Reference	Difference (pp)
Huginn-3.5B	ARC	77.06 [75.31, 78.72]	45.37	31.69
	MMLU	58.01 [57.18, 58.84]	44.32	13.69
RL-1.4B (R32)	ARC	67.51 [64.95, 69.82]	39.25	28.26
	MMLU	16.94 [15.98, 17.88]	22.35	-5.41

Shared question-level variation can contribute to the covariance sum. Neither zero excess nor a specified sign establishes coordinate independence. This reference describes empirical joint structure without assuming exchangeability under a generative model. The raw-radius-weighted removable fraction τ (Equation 18) has no corresponding quadratic covariance identity.

6 Sensitivity to Scoring and Numerical Precision

The preceding identities characterize a fixed score trajectory; the observations also depend on how that trajectory is obtained. We examine this dependence through paired comparisons of scoring construction and numerical precision, with each scoring format retaining its own endpoint. Uniform rescaling then identifies which statistics are invariant to score units.

6.1 Scoring construction and endpoint predictions

For Huginn, the paired readouts produce different stability profiles and endpoint answers: answer-text scoring has lower disagreement with its own endpoint, while the two readouts often select different endpoint answers. This comparison changes the prompt and execution as well as the scoring formula. On the $N = 32$ questions per task shared by these conditions, let $y_{i,t}^{(f)}$ be the answer identity under readout $f \in \{\text{lab}, \text{txt}\}$, where label scoring displays the options and answer-text scoring omits the options block. With $T = 32$ and $\mathcal{T} = \{8, 16, 24\}$, define

$$\begin{aligned} \varepsilon_f &= \frac{1}{N|\mathcal{T}|} \sum_{i=1}^N \sum_{t \in \mathcal{T}} \mathbf{1}\{y_{i,t}^{(f)} \neq y_{i,T}^{(f)}\}, & A_f &= \frac{1}{N} \sum_i \mathbf{1}\{y_{i,T}^{(f)} = y_i^*\}, \\ a_{\text{lab}, \text{txt}} &= \frac{1}{N} \sum_i \mathbf{1}\{y_{i,T}^{(\text{lab})} = y_{i,T}^{(\text{txt})}\}. \end{aligned} \quad (34)$$

Here ε_f is the fraction of sampled intermediate answers that disagree with their own endpoint, A_f is endpoint accuracy against the ground-truth answer y_i^* , and $a_{\text{lab}, \text{txt}}$ is endpoint agreement between the two readouts. The paired text-minus-label differences in ε_f are -28.13 pp $[-39.58, -17.71]$ on ARC and -29.17 pp $[-41.67, -16.67]$ on MMLU; $a_{\text{lab}, \text{txt}} = 10/32$ and $12/32$, respectively. Table 12 also reports the normalized earliest-depth areas W_j for raw, mean-centered, and quotient tests.

Table 12: Same-question prediction and geometry: Huginn-3.5B (0125), $N = 32/\text{task}$, $T = 32$, candidate depths $\{8, 16, 24\}$. F1 denotes label scoring and F2 answer-text scoring without displayed options (Equation 14). Disagreement averages $\mathbf{1}\{y_t \neq y_{32}\}$; the last columns give W (%). Blue marks paired disagreement reductions versus label scoring; mint marks quotient gains over mean centering, each with 95% intervals excluding zero.

Task	Format	Correct / N	Disagreement	Raw	Mean	Quotient
ARC	F1	8/32	34.38	18.75	29.69	32.81
	F2	14/32	6.25	42.97	53.12	57.03
MMLU	F1	10/32	36.46	19.53	24.22	25.78
	F2	15/32	7.29	34.38	45.31	50.78

Under answer-text scoring, $W_q - W_{\text{raw}}$ is 14.06 pp on ARC and 16.41 pp on MMLU; $W_q - W_{\text{mean}}$ is 3.91 $[0.78, 8.59]$ and 5.47 $[0.78, 11.72]$ pp, respectively. Both quotient-over-mean intervals exclude zero, whereas both paired accuracy intervals for $A_{\text{txt}} - A_{\text{lab}}$ include zero. Lower endpoint disagreement therefore does not by itself imply an accuracy gain. The sum and token-normalized scores below complete the comparison, while Figure 7 shows the prediction profiles.

Score normalization. Using the character-normalized F2 score in Equation 14, the alternative normalizations are

$$s_{t,k}^{(L)} = \frac{\ell_k}{L_k} s_{t,k}^{\text{txt}}, \quad L_k \in \{1, n_k, \ell_k\}, \quad (35)$$

where $L_k = 1, n_k, \ell_k$ gives the sum, token mean, and character mean, respectively, on the same answer continuations. Character mean is the main F1/F2 comparison; Table 13 retains all three variants. Actual target-token log probabilities supply the sums: later-position columns for the first candidate token do not supply those targets. Reconstructed scores differ from stored reductions by at most 1.01×10^{-5} , with identical selected answers.

Endpoint-conditioned comparisons. The paired text-minus-label accuracy differences are 18.75 $[-3.12, 40.62]$ and 15.62 $[-3.12, 34.38]$ pp. Conditioning on each format’s own correct endpoint gives quotient-over-raw depth increments of 9.38 $[3.12, 18.75]$ pp for ARC/F1 ($n = 8$), 17.86 $[10.71, 25.00]$ for ARC/F2 ($n = 14$), 7.50 $[0.00, 15.00]$ for MMLU/F1 ($n = 10$), and 15.00 $[8.33, 23.33]$ for MMLU/F2 ($n = 15$). Only four ARC and seven MMLU questions belong to both correct-endpoint subsets. On the paired eight-question precision panels, F1/F2-char depth increments change from 12.50/9.38 to 18.75/18.75 pp on ARC and from 12.50/18.75 to 12.50/21.88 pp on MMLU between BF16 and FP32 recurrence. All depth quantities normalize recurrence within a scoring format; F2 executes multiple answer continuations whereas F1 uses one shared prefix.

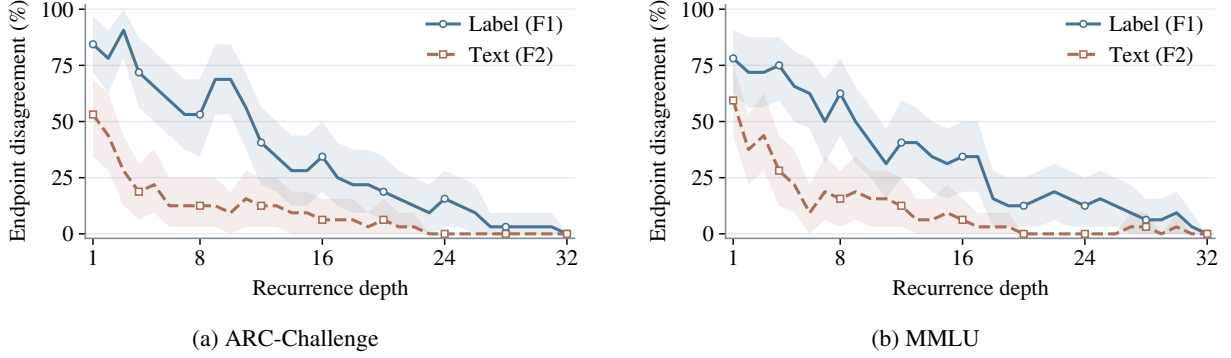


Figure 7: Distinct prediction trajectories under label and answer-text scoring. Huginn-3.5B (0125) on the same 32 questions per task, with label scoring (F1) and character-normalized answer-text scoring (F2). Each curve compares depth t with its own depth-32 winner; shading shows pointwise 95% whole-question bootstrap intervals. The scoring formulas are defined in Section 4; Table 12 gives the paired comparison.

Table 13: All population scoring variants on the same 32 questions per task in Huginn-3.5B (0125). Ξ and τ are the common-energy and removable-radius shares (Equations 32 and 18), computed as adjacent-increment ratios of sums, in percent. $\Delta W = W_q - W_{\text{raw}}$ is in percentage points, relative to each format’s own $T = 32$ endpoint; brackets give 95% paired intervals.

Task	Score	Correct	Ξ	τ	ΔW : 8, 16, 24	ΔW : 1:31
ARC	F1	8	59.65	41.19	14.06 [7.81, 21.09]	16.41 [10.84, 23.05]
	F2 char	14	71.05	52.42	14.06 [9.38, 19.53]	21.58 [15.82, 27.64]
	F2 sum	9	72.04	52.14	17.19 [10.16, 25.00]	21.19 [15.33, 27.34]
	F2 token	11	70.92	52.08	11.72 [6.25, 17.19]	18.07 [12.89, 23.34]
MMLU	F1	10	60.65	44.09	6.25 [2.34, 10.94]	20.12 [12.01, 29.00]
	F2 char	15	80.43	53.47	16.41 [9.38, 24.22]	23.14 [15.92, 30.86]
	F2 sum	16	91.73	56.57	21.09 [14.06, 29.69]	22.75 [15.23, 31.35]
	F2 token	13	77.95	53.15	17.97 [10.94, 25.00]	18.85 [11.33, 26.96]

6.2 Numerical precision and token composition

The archived-format precision study evaluates the same 32 questions per task in both Raven-lineage checkpoints. The three arms use native BF16, an FP32 head applied to the same BF16 hidden values, and full-FP32 recurrence with the same BF16-rounded weights and saved initial tensors. These arithmetic changes can affect translation identities in finite precision (Blanchard et al., 2021).

We use the retained-radius ratio $\rho_{i,t}$ and removable fraction τ defined in Equation 18. Thus τ is a raw-radius-weighted removable fraction, whereas the atom at zero in Figure 8(a,b) counts exact nonzero translations.

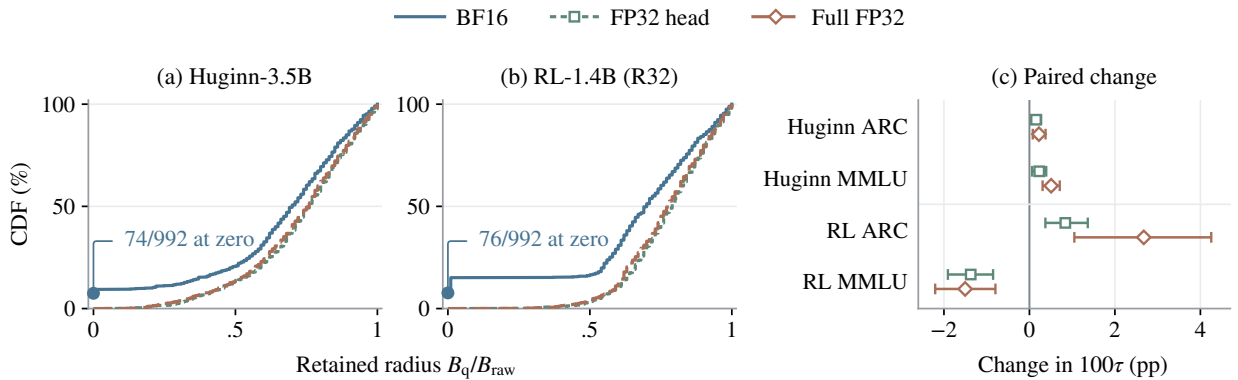


Figure 8: Exact plateaus disappear while aggregate translation persists. (a,b) MMLU distributions of B_q/B_{raw} over nonzero adjacent increments, sampled on a 0.01 grid; an atom at zero denotes exact translation. (c) Paired changes in raw-radius-weighted 100τ , with 95% whole-question intervals. Each checkpoint has the same 32 questions per task across arms. FP32-head preserves BF16 hidden states; full FP32 changes upstream arithmetic with the same rounded weights.

Table 14 gives the aggregate removable-radius shares and 95% intervals from 2,000 paired whole-question resamples. Native MMLU contains 74/992 exact zero-span, nonzero-row increments in Huginn and 76/992 in Recurrent-Llama. Both counts become zero in each higher-precision arm, while aggregate translation removal remains substantial with model-dependent changes. Thus exact candidate-contrast plateaus depend on arithmetic precision; these comparisons do not establish precision invariance of every full-text depth area or mass-concentration result. Figure 9 completes the retained-radius distributions for ARC under the same plotting definitions.

Table 14: Adjacent-increment radius reduction under paired precision changes in the archived scoring formats. Values are percentages with 95% intervals. H: Huginn-3.5B (0125); RL: Recurrent-Llama-1.4B (R32).

Model	Task	BF16	FP32 head	Full FP32
H	ARC	51.88 [49.13, 54.57]	52.04 [49.23, 54.76]	52.11 [49.29, 54.89]
H	MMLU	44.34 [42.57, 46.02]	44.57 [42.75, 46.33]	44.84 [42.98, 46.63]
RL	ARC	50.08 [45.67, 54.90]	50.92 [46.23, 56.07]	52.76 [47.14, 58.72]
RL	MMLU	24.66 [23.12, 26.11]	23.28 [21.82, 24.71]	23.15 [21.42, 24.86]

For ARC, split each score into its first-token and remaining-token terms, and denote their cross-choice mean increments by u and v . The identity $K(u+v)^2 = Ku^2 + Kv^2 + 2Kuv$ yields first/later/cross contributions of 46.38/35.63/17.99% for Huginn and 44.26/27.77/27.97% for Recurrent-Llama at native precision. The signed cross term retains the interaction between token contributions. Matched prefix-only executions also show numerical shape sensitivity: the maximum vocabulary-log-probability discrepancy falls from 0.372150 to 5.8×10^{-5} for Huginn, and from 0.254886 to 3.7×10^{-5} for Recurrent-Llama, between BF16 and full FP32. The pre-head hidden values change as well.

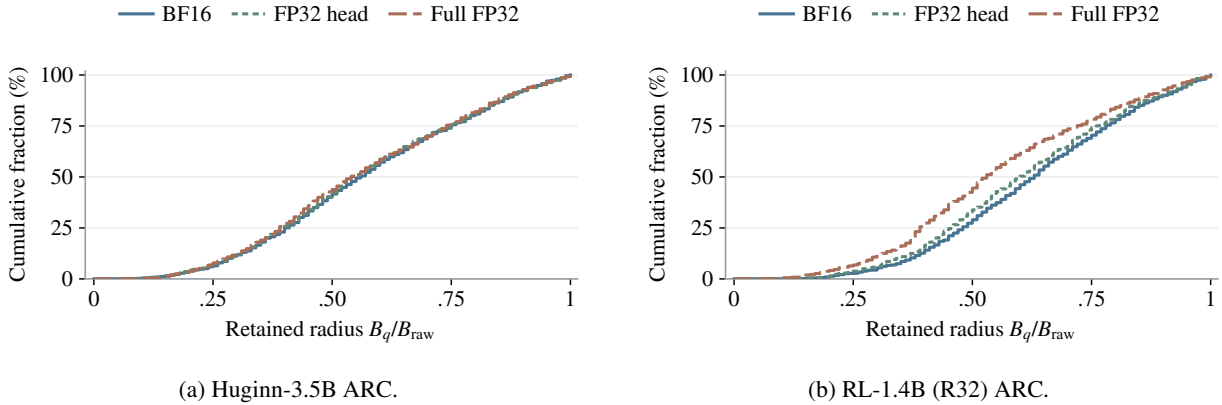


Figure 9: Precision-dependent retained-radius distributions on ARC. Each checkpoint uses the same 32 questions across all arithmetic arms; ratios are sampled on the same 0.01 grid as Figure 8(a,b). The distribution weights each nonzero adjacent increment equally, whereas τ weights removable fractions by the corresponding raw radius.

6.3 Score scale and centering

Uniform positive rescaling tests which properties depend on score units. For $\alpha > 0$, every selected answer is preserved. Raw, mean-centered and quotient radii, directed bounds, reserves, and margins scale by α . On a fixed candidate-depth set,

$$\begin{aligned} (Q, C)(\alpha\delta) &= \alpha^2(Q, C)(\delta), \\ D_j(\alpha s) &= D_j(s), \quad j \in \{\text{raw}, \text{mean}, \text{q}, \text{dir}, \text{res}\}. \end{aligned} \quad (36)$$

For $Q + C > 0$, the contrast share $Q/(Q + C)$ is therefore invariant. The same holds for τ when its raw-radius sum is positive and Ξ when its total energy is positive. LSE centering is not homogeneous: in general, $\text{LSE}(\alpha s_T) - \text{LSE}(\alpha s_t) \neq \alpha(\text{LSE}(s_T) - \text{LSE}(s_t))$.

The LSE representative in the original score units is $a_\alpha(s) = \alpha^{-1} \text{LSE}(\alpha s)$ (Gao and Pavel, 2017). For a fixed finite score vector and the uniform distribution U_K on its K coordinates, as $\alpha \rightarrow 0^+$,

$$a_\alpha(s) - \frac{\log K}{\alpha} = \bar{s} + \frac{\alpha}{2} \text{Var}_{U_K}(s) + O(\alpha^2). \quad (37)$$

At fixed K across depths, the divergent $\log K/\alpha$ terms cancel, so $\Delta a_\alpha(s) \rightarrow \Delta \bar{s}$. The following derivation gives an explicit remainder bound for the expansion.

Proof of Equation 37. Fix $s \in \mathbb{R}^K$ and write

$$f(u) = \log \left(\frac{1}{K} \sum_{k=1}^K e^{us_k} \right), \quad \pi_{u,k} = \frac{e^{us_k}}{\sum_l e^{us_l}}, \quad L = \text{osc}(s).$$

The function f is smooth on $\mathbb{R}$, with

$$\begin{aligned} f(0) &= 0, & f'(u) &= \mathbb{E}_{\pi_u} s, \\ f''(u) &= \text{Var}_{\pi_u}(s), & f^{(3)}(u) &= \mathbb{E}_{\pi_u}(s - \mathbb{E}_{\pi_u} s)^3. \end{aligned}$$

Since $\mathbb{E}_{\pi_u} s \in [\min_k s_k, \max_k s_k]$,

$$|f^{(3)}(u)| \leq \mathbb{E}_{\pi_u} |s - \mathbb{E}_{\pi_u} s|^3 \leq L^3.$$

For $\alpha > 0$, Taylor’s formula with integral remainder yields

$$f(\alpha) = \alpha \bar{s} + \frac{\alpha^2}{2} \text{Var}_{U_K}(s) + \frac{1}{2} \int_0^\alpha (\alpha - u)^2 f^{(3)}(u) du.$$

Using $a_\alpha(s) = (\log K + f(\alpha))/\alpha$,

$$\begin{aligned} \left| a_\alpha(s) - \frac{\log K}{\alpha} - \bar{s} - \frac{\alpha}{2} \text{Var}_{U_K}(s) \right| &\leq \frac{L^3}{2\alpha} \int_0^\alpha (\alpha - u)^2 du \\ &= \frac{\alpha^2 L^3}{6}. \end{aligned} \tag{38}$$

□

For fixed K , the $\log K/\alpha$ terms cancel in a two-depth difference, and the two remainder bounds add. This gives the $O(\alpha^2)$ expansion as $\alpha \rightarrow 0^+$, uniformly over score vectors with bounded span.

Across the native population, scales $\alpha \in \{0.1, 0.25, 0.5, 1, 2, 4, 10\}$ preserve all winners and the homogeneous statistics. With $b_\alpha = \Delta a_\alpha(s)$, the aggregate excess over the quotient radius after LSE centering, normalized by the original raw radii, is $\sum 2|\text{mid}(d) - b_\alpha| / \sum 2\|d\|_\infty$ for a positive denominator (Equation 2). It changes from 14.99% at $\alpha = 0.1$ to 21.87% at $\alpha = 1$ and 50.89% at $\alpha = 10$ for Recurrent-Llama MMLU. Its mean-centered counterpart remains 15.08%. Mean centering has less residual than LSE centering in all four model/task cells at native scale. Thus an LSE residual depends on the chosen score scale; its set-mass interpretation additionally requires a shared predictive distribution (Section 5.1).

7 From Descriptive Geometry to Prediction

The preceding analyses use completed trajectories to characterize score changes and answer preservation. We now examine what the readout geometry predicts on held-out questions: first the second moments and common-energy allocation of updates, then endpoint agreement from an observed prefix. The fitted comparisons retain their training and evaluation split; the prefix study uses its own previously specified policy evaluation.

7.1 Predicting readout moments with matched energy

The moment and allocation studies use the previously inspected Huginn paired-format population of 32 questions per task (Appendix A.1). Question-level folds separate fitting from evaluation within this population. We compare an independent-logit innovation reference (N1) with a reference that applies the model’s actual output head to isotropic hidden innovations (N2). Linear propagation and direct nonlinear readout evaluation use the same observation map. Four question-level folds contain 24 training and eight held-out questions per task; all depths and representations of a question share its fold. Nonlinear evaluation uses 512 bounded Rademacher draws at the five fixed transition anchors 4, 8, 16, 24, 31.

For the following moment and allocation comparisons, $y \in \mathbb{R}^K$ denotes an observed score increment. For predicted mean μ_j and positive-semidefinite covariance C_j with $\mathbb{E}_{\text{train}} \text{tr}(C_j) > 0$, training-energy matching sets

$$\alpha_j = \frac{\mathbb{E}_{\text{train}} \|y - \mu_j\|^2}{\mathbb{E}_{\text{train}} \text{tr}(C_j)}, \quad \tilde{C}_j = \alpha_j C_j. \tag{39}$$

The means remain fixed. Losses compare predicted raw second moments with the observed increment outer products, using the training normalization and averaging depths within questions. Table 15 reports the comparisons after matching training energy. Linear N2 has higher loss than N1 on ARC/F1; their MMLU/F1 difference includes zero. Direct nonlinear evaluation improves ARC/F1 relative to linear N2; its comparison with matched N1 has an interval extending to zero. F2 here denotes unnormalized text-score sums.

Table 15: Paired raw-second-moment loss differences after training-energy matching. Negative values favor the first-named method. Brackets are 95% intervals conditional on the fitted models.

Task	Score	Linear N2 – N1	Nonlinear N2 – N1	Nonlinear – linear N2
ARC	F1	0.1921 [0.0925, 0.2991]	-1.2011 [-2.3992, 0.0046]	-1.3932 [-2.6346, -0.1480]
MMLU	F1	0.0664 [-0.0258, 0.1820]	0.0882 [-1.1059, 1.3802]	0.0218 [-1.2305, 1.3868]
ARC	F2 sum	-0.0513 [-0.1099, 0.0106]	-0.0873 [-0.4275, 0.1502]	-0.0361 [-0.4178, 0.2301]
MMLU	F2 sum	0.0706 [-0.0141, 0.1822]	-2.3638 [-7.2797, 0.2055]	-2.4344 [-7.4285, 0.1509]

7.2 Predicting allocation after matching energy

To separate allocation from total scale, set $e = y - \mu$ using the same training empirical mean for every method, $u = \mathbf{1}/\sqrt{K}$, $A = (u^\top e)^2$, and $B = \|e\|^2 - A \geq 0$.

Let $p \in [0, 1]$ predict the common share of residual energy. With training residual energy $E_0 > 0$, the weighted loss

$$\ell(p; A, B) = \frac{A + B}{E_0} \left(p - \frac{A}{A + B} \right)^2 \quad (40)$$

is defined to be zero when $A + B = 0$.

Proof of optimal allocation under the weighted loss. Condition on the fitted training quantities, including $E_0 > 0$. For a fixed x , write

$$S = A + B, \quad s_x = \mathbb{E}[S \mid x], \quad a_x = \mathbb{E}[A \mid x].$$

Assume $0 < s_x < \infty$, and define $A^2/S = 0$ on $\{S = 0\}$. Since $0 \leq A \leq S$ and $0 \leq A^2/S \leq S$,

$$0 \leq a_x \leq s_x, \quad c_x := \mathbb{E}[A^2/S \mid x] < \infty.$$

For $p \in \mathbb{R}$, expansion of Equation 40 gives

$$R_x(p) := \mathbb{E}[\ell(p; A, B) \mid x] = \frac{s_x p^2 - 2a_x p + c_x}{E_0}.$$

Setting $p^* = a_x/s_x \in [0, 1]$ yields

$$R_x(p) - R_x(p^*) = \frac{s_x}{E_0} (p - p^*)^2.$$

As $s_x/E_0 > 0$, p^* is the unique minimizer on both $\mathbb{R}$ and $[0, 1]$. If $s_x = 0$, then $S = 0$ conditionally almost surely, so $R_x(p) = 0$ for every p . $\square$

The empirical comparator uses the training ratio of sums. A two-parameter ridge predictor uses the standardized linear-N2 allocation as its sole feature, with unit slope penalty and unpenalized intercept. Table 16 reports the direct nonlinear prediction and ridge against the same constant. The earlier two-block loss uses a different objective,

$$\ell_2(p; A, B) = (p - A/E_0)^2 + (1 - p - B/E_0)^2.$$

Proof of the two-block minimizer. Assume $\mathbb{E}[A^2 + B^2 \mid x] < \infty$ and define

$$p_0 = \frac{1}{2} + \frac{\mathbb{E}[A - B \mid x]}{2E_0}.$$

Direct expansion gives, for every $p \in \mathbb{R}$,

$$\mathbb{E}[\ell_2(p; A, B) - \ell_2(p_0; A, B) \mid x] = 2(p - p_0)^2.$$

Consequently,

$$\arg \min_{p \in \mathbb{R}} \mathbb{E}[\ell_2(p; A, B) \mid x] = \{p_0\}, \quad \arg \min_{p \in [0, 1]} \mathbb{E}[\ell_2(p; A, B) \mid x] = \{\Pi_{[0, 1]} p_0\}.$$

$\square$

All fitted predictions lie in $[0, 1]$. The earlier loss gave an MMLU/F2 ridge difference of $-0.1029 [-0.2699, -0.0078]$. Under Equation 40, the corresponding weighted-ridge interval extends to zero. All predictive intervals use 2,000 paired question resamples with fits and simulated draws held fixed.

Table 16: Energy-allocation loss relative to the training constant under Equation 40. Negative differences favor the candidate; the four main task/format conditions are shown.

Task	Score	Nonlinear geometry – constant	Weighted ridge – constant
ARC	F1	0.05625 [0.02391, 0.08962]	−0.01231 [−0.03936, 0.00810]
ARC	F2 sum	0.09697 [0.04030, 0.15908]	0.00369 [−0.00844, 0.01666]
MMLU	F1	0.06529 [0.02726, 0.10676]	−0.00231 [−0.01554, 0.00991]
MMLU	F2 sum	0.13603 [0.01379, 0.28691]	−0.03744 [−0.10122, 0.00030]

7.3 Predicting endpoint agreement from a prefix

Completed-trajectory geometry and prefix prediction can also be compared empirically. An earlier nested, trajectory-held-out prefix study on the native Huginn collection uses $T = 32$, a four-step observation window and candidate depths 4:31. At the 2.5% target disagreement level, normalized work reductions are 21.72% for raw, 23.93% for mean-centered and 22.97% for quotient ranking, with empirical disagreement among exits of 2.46, 2.54 and 2.54%, respectively. The paired quotient difference is 1.26 [1.08, 1.43] pp against raw and −0.95 [−1.10, −0.82] pp against mean; a train-selected richer comparator is ahead by 1.36 [1.01, 1.71] pp. These earlier policy results concern observed-prefix rankings rather than the realized increments used throughout the geometric analysis.

8 Discussion

The two decompositions resolve different aspects of a prediction trajectory. Common and contrast components describe what the score update contains; the shared-distribution factorization gives common motion a probability interpretation through mass and concentration. Winner-directed motion and competitor gaps determine whether the update crosses a decision boundary. Consequently, neither update magnitude nor an energy share alone specifies its decision consequence. The measured direction and pairing contributions under label and answer-text scoring show that this distinction accounts for a substantial part of the gap between magnitude bounds and realized answer preservation.

The statistical references further distinguish mathematical structure from predictive performance. In the matched-label comparison, actual Huginn updates preserve the winner less often than the exact displacement-coordinate permutation reference. In the native-precision archive, late MMLU common energy falls below its coordinate-marginal expectation in both checkpoints. After energy matching, direct readout geometry does not outperform the constant allocation reference in the four reported conditions. These results constrain what can be inferred from a large common component or a favorable retrospective margin: neither alone identifies a predictive model of future updates. In the prefix study, quotient ranking improves on the raw comparator but remains behind mean centering and the richer comparator selected on training data. Retrospective geometric gains therefore need a separate predictive evaluation before they can inform a stopping policy.

The study covers multiple-choice predictions from three checkpoints in two model families. Completed-trajectory depth areas describe geometric headroom; an online policy would additionally need to predict future relative updates from a prefix. Interventions on recurrent states could identify the neural processes that generate the mass and concentration terms. Extending the analysis to open-ended generation requires accounting for changing candidate sets and the interaction between recurrent depth and generated context.

Reproducibility Statement

Section 4 defines the scoring formulas. Appendix A specifies fixed model revisions, endpoints, precision settings, resampling units, and question selection and overlap. Mathematical identities are proved beside their statements. Each empirical analysis reports its population, comparisons and uncertainty intervals; predictive analyses in Section 7 retain their fitted training quantities and held-out evaluation roles.

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A Experimental Design

The preceding analyses use complete trajectories under fixed readout and endpoint choices. We collect here the model configurations, question populations and resampling procedures needed to reproduce those comparisons. Score constructions are defined in Section 4; each result identifies its applicable population. The distinctions between previously inspected questions and the separate new-question population remain essential to interpreting the empirical evidence.

A.1 Checkpoints and trajectory populations

We evaluate Huginn-3.5B, Recurrent-Llama-1.4B and Ouro-1.4B on ARC-Challenge (Clark et al., 2018) and MMLU (Hendrycks et al., 2021), with fixed evaluation rules (Biderman et al., 2024). Huginn and Recurrent-Llama have distinct weights and training runs within the Raven lineage; Ouro supplies a second model family. Parameter counts refer to shared weights, separately from recurrence depth: Huginn’s official 3.5B size counts unique parameters; Recurrent-Llama has 1,385,228,288 parameters after its Llama-3.2-1B retrofit (McLeish et al., 2025), including untied embeddings, with R32 denoting its training recurrence. Ouro’s 1.4B shared parameters are independent of its four-loop inference horizon. Tables A.1 and A.2 give the collections and pinned revisions.

Table A.1: Trajectory collections; N is the number of questions per task. H: Huginn-3.5B; RL: Recurrent-Llama-1.4B; O: Ouro-1.4B. Label, text without options and text with options correspond to F1, F2 and F3. H/O questions are paired for label and displayed-text scoring; H label/text comparisons use the same questions.

Collection	Models	N	T	Scoring condition
Native trajectories	H, RL	1,352	32	Native evaluator
Matched labels	H, O	128	32, 4	Label; options displayed
Paired label/text	H	32	32	Label / text without options
Text, four arrangements	H, O	32	32, 4	Text; options displayed
Precision comparison	H, RL	32	32	Native evaluator

Table A.2: Pinned model configuration. Parameter counts refer to shared weights, not their repeated use across depths. The Huginn/Ouro matched-label and answer-text collections use BF16 recurrence with an FP32 output head; the Huginn/Recurrent-Llama archive uses native BF16. Paired arithmetic arms are specified in Section 6.2.

Model / checkpoint	Pinned revision	Parameters	Evaluated T
Huginn-3.5B (0125)	bb6621b65e90	3.5B	32
Recurrent-Llama-1.4B (R32)	a5f6f126e9d9	1.4B	32
Ouro-1.4B	574fa66cb8bf	1.4B	4

Archived and paired-format populations. The archive contains 2,704 questions, equally divided between ARC-Challenge and MMLU. Huginn and Recurrent-Llama each contribute a complete 32-depth trajectory for every question: 86,528 readout vectors and 83,824 adjacent increments per checkpoint. Archived ARC scores sum continuation log probabilities and divide by character length; each MMLU continuation contains one answer token. Four ARC questions have three choices and two have five; all remaining questions have four. Coordinate permutations are stratified by choice count, task and depth. A fixed hash selects 32 questions per task from this archive for the paired-format study. Huginn evaluates each in F1 and F2 over all 32 depths; its paired full-FP32 subset contains eight questions per task. These previously inspected data support the readout-reference and same-question format comparisons, rather than a new-question confirmation.

Separate matched-label depth population. The depth study uses 128 additional questions per task, absent from the project’s preceding question manifests. Its early-late prediction was specified before collecting the Huginn outputs. Ouro evaluates the same questions with identical F1 prompt text at its native four-loop horizon; tokenization remains model-specific. A fixed 16-question subset per task and model supplies the paired precision analysis.

Full answer-text geometry on reused questions. This collection reuses the same 32 previously inspected questions per task as the paired-format study. Huginn and Ouro score every complete answer with character normalization and all options displayed (F3). Four fixed arrangements, (1, 2, 3, 4), (2, 4, 1, 3), (4, 3, 2, 1) and (3, 1, 4, 2), balance each answer across positions while retaining each continuation’s semantic identity. Every question contributes, including those with wrong endpoints. The primary analysis applies an FP32 output head to each complete sequence after BF16

recurrence, uses native endpoints $T = 32/4$ and the quarter-depth grid, and takes the output at each recurrent depth without Ouro’s gate mixture. The readout formula, arrangements, depth grid and comparisons were fixed before the Ouro collection. This extends the analysis to another model family on reused questions without changing their data role.

Unless a precision comparison is specified, matched-label and text runs use BF16 recurrence with an FP32 output head; archived native trajectories use the native evaluator’s BF16 arithmetic. Section 6.2 specifies the paired arithmetic arms, including the separate native-evaluator comparison on 32 questions per task and checkpoint.

A.2 Paired uncertainty estimates

We resample whole questions, retaining every depth, scoring variant and paired condition of each sampled question. Unless specified otherwise, intervals are marginal 95% percentile bootstrap intervals from 5,000 draws. Archive and format comparisons describe previously inspected data; the fixed new-question depth comparison uses the separate 128-question collection. Precision and auxiliary prediction comparisons use 2,000 paired draws, with fitted quantities held fixed for the latter.